McMaster University

Department of Mathematics and Statistics

Accredited Courses that may be used towards the A.Stat. designation

MODULE	ACCREDITED COURSE
Mathematics Modules	
1. Calculus 1 (1 full-year course required) *Either MATH 1A03 and MATH 1AA3* *or MATH 1X03 and MATH 1XX3*	MATH 1A03: Calculus for Science I; or MATH 1AA3: Calculus for Science II; or MATH 1X03: Calculus for for Math and Stats I; or MATH 1XX3: Calculus for Math and Stats II
2. Calculus 2 (1 course required)	MATH 2X03: Advanced Calculus I
3. Linear Algebra (1 course required)	MATH 1B03: Linear Algebra I
Statistics and Probability Modules	
4. Mathematical Statistics (1 full-year course required) *STATS 3D03 and either STATS 4CI3 or STATS 3DD3*	STATS 3D03: Mathematical Statistics STATS 4CI3: Computational Methods for Inference (replaced STATS 3DD3 in 2009)
5. Linear Regression (1 course required)	STATS 3A03: Applied Regression Analysis with SAS
6. Survey Sampling (Required if Experimental Designs not taken)	STATS 3S03: Survey Sampling
7. Statistics Elective (3 courses required) *Any 3 courses from the following list. STATS 4P03 may not be used for this module if it is used for the "Communications Skills" or "Substantive Area" modules. STATS 3G03, 3H03, 3PG03 or 4D03 may not be used for this module if used for the "Substantive Area" module.* STATS 3F03: Categorical Data Analysis STATS 3G03: Actuarial Mathematics I STATS 3H03: Actuarial Mathematics II STATS 3PG3: Probability of Games of Chance STATS 3U03: Stochastic Processes STATS 4A03: Time Series STATS 4C03: Generalized Linear Models STATS 4D03: Intermediate Probability STATS 4M03: Multivariate Analysis STATS 4P03: Advanced Applied Statistics	

8. Computer Skills (2 courses required) *The courses in the following list each have a computer skills component; completion of any 2 of them will satisfy the "Computer Skills" requirement.* STATS 2MB3: Statistical Methods and Applications [R] STATS 3A03: Applied Regression Analysis with SAS [SAS] STATS 4C03: Generalized Linear Models [R] STATS 4CI3: Computational Methods for Inference [R] STATS 4M03: Multivariate Analysis [R] STATS 4P03: Advanced Applied Statistics [R, SAS] MATH 2T03: Numerical Analysis I [MATLAB]
9. Communications Skills (1 course required) STATS 4P03: Advanced Applied Statistics
10. Substantive Area (3 courses required) *Any 3 courses chosen from the following list and from 3rd and 4th year level courses within or outside the Department of Mathematics & Statistics with some application content. Also, documented 4-month co-op placements are acceptable.* STATS 3G03: Actuarial Mathematics I STATS 3H03: Actuarial Mathematics II STATS 3PG3: Probability and Games of Chance STATS 4D03: Intermediate Probability STATS 4P03: Advanced Applied Statistics MATH 2FM3: Introduction to Mathematical Finance MATH 3FM3: Mathematical Finance MATH 4FM3: Financial Markets and Derivatives MATH 4MB3: Mathematical Biology

Course Outlines

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Math 1A03/1ZA3 (Fall 2015) Course Outline

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Course Home Page

- The course home page is NOT on Avenue to Learn. It is accessible from the link at the following web site: http://ms.mcmaster.ca/childsa/1za3.html

Course Description

- **Course Title:** Math 1A03/1ZA3 - Calculus for the Physical Sciences I/Engineering Mathematics I
- **Class Times and Locations:** Check on Mosaic.

Math 1A03 Section 1 (C01) Instructor Information

- **Name:** David Duncan
- **email:** duncand@math.mcmaster.ca
- **Office Location:** HH/319
- **Office Hours:** Tuesday and Wednesday 10:30am-11:30am

Math 1A03 Section 2 (C02) Instructor Information

- **Name:** Federico Galetto
- **email:** galettof@math.mcmaster.ca
- **Office Location:** HH/407
- **Office Hours:** Wednesday 1:30pm-4:00pm (or by appointment)

Math 1ZA3 Section 1 (C01) Instructor Information

- **Name:** Chris McLean
- **email:** mcleac3@math.mcmaster.ca
- **Office Location:** BSB/B124
- **Office Hours:** Tuesday, Thursday, and Friday 10:00am-11:30am

Math 1ZA3 Section 2 (C02) Instructor Information

- **Name:** Aaron Childs
- **email:** childsa@mcmaster.ca
- **Office Location:** HH/213
- **Office Hours:** Click here

Math 1ZA3 Section 3 (C03) Instructor Information

- **Name:** Jean Pierre Gabardo
- **email:** gabardo@mcmaster.ca
- **Office Location:** HH/309
- **Office Hours:** Wednesday 2:00pm-3:00pm, Thursday 1:00pm-2:00pm, or by appointment (e-mail preferred)

Textbook

- **Required:**
  -*Calculus, Early Transcendentals, 8th Edition*, James Stewart, Brooks/Cole.
  (**Note:** Older editions can be used, as long as you have access to the exercises in the 8th edition.)
- **Optional:**
  -Calclabs with Maple for Single Variable Calculus
  -Calclabs with Maple for Multivariable Calculus
  (or Calclabs with Maple Custom Edition, which includes the sections that you will need from both of the above Calclabs manuals)

- Student Solutions Manual for Single Variable Calculus, Early Transcendentals
- Student Solutions Manual for Multivariable Calculus
- Maple 11 (or later) Software (earlier versions of Maple cannot be used)

- A copy of the textbook and solutions manual are available on reserve in Thode Library.

Material Covered

- All Sections covered in the Suggested Problems
- **Major Topics:** Continuity and differentiability, with emphasis on theory (intermediate value theorem, mean value theorem), practice (how to differentiate) and applications (curve sketching, optimization), theory and techniques of integration, with emphasis on practice (how to integrate) and applications.

Assignment Information

- There will be 6 online assignments. See the calendar for due dates.

Lab Information

- There will be 5 labs which will require the use of Maple (Version 11 or later) and will be completed electronically using our online system. See the calendar for due dates.
- You do not have to attend any scheduled lab times. But TAs will be available if you need help at the times given on the Lab information page.
- All information about labs is available on the Lab information page.

Test Information

- Only the McMaster standard calculator Casio fx-991 is allowed on the tests and exam
- Some sample tests are available on the Suggested Problems pages
- **Tentative Dates** (subject to change):
  Test #1: Evening of Monday October 19th (75 minutes)
  Test #2: Evening of Monday November 16th (75 minutes)
- Check the Announcements part of the course web site for room and time information, and for instructions on what to do if you have a conflict with the test time

Course Evaluation

6 Assignments - 2% each
5 Labs - 2% each
2 Tests - 19% each
Final Exam - 40%

Notes:

- At the end of the course the grades may be adjusted, but this can only increase your grade and will be done uniformly. We will use the grade equivalence chart published in the Undergraduate Calendar to convert between percentages and letter grades.
- The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Missed Work Policy

- If you are absent from the university for a minor medical reason, lasting fewer than 3 days, you may report your absence, once per term, without documentation, using the McMaster Student Absence Form. Absences for a longer duration or for other reasons must be reported to your Faculty/Program office, with documentation, and relief from term work may not necessarily be granted. When using the MSAF, report your absence to childsa@mcmaster.ca. Please note that the MSAF may not be used for term work worth 25% or more, nor can it be used for the final examination. For more information look here.
- If your MSAF form was received then the word "note" will appear in place of your mark on the online system. This will show up within one week after you filled out the MSAF form. If you don't see the word "note" in place of your mark for the missed work one week after filling out the MSAF form, then send an email to Dr. Childs telling him the date that you filled out the MSAF form. If you do see the word "note" in place of your mark, then no follow-up is required.
- The perentage for a missed test will be added to your final exam.
- The percentage for a missed assignment or lab will be distributed among your remaining assignments or labs.

Academic Dishonesty Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3, located at http://www.mcmaster.ca/policy/Students-AcademicStudies/AcademicIntegrity.pdf The following illustrates only three forms of academic dishonesty: 1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained. 2. Improper collaboration in group work. 3. Copying or using unauthorized aids in tests and examinations.

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Mathematics 1AA3/1ZB3 - Term 2, Winter 2016

Updated Jan 6, 2016

Download PDF version

Instructors:

Course:	Section:	Lecturer:	Office	Phone Ex.	E-Mail:
1AA3	C01	A. Nicas	HH 310	23427	nicas@mcmaster.ca
1ZB3	C01	C. McLean	BSB B124	20142	mcleac3@mcmaster.ca
1ZB3	C02	J.P. Gabardo	HH 309	23416	gabardo@mcmaster.ca
1ZB3	C03	J. Feys	HH 327	23413	feysj@math.mcmaster.ca

Contacting the Instructors:

E-mails are the preferred method of communication. All inquiries should either be done in person after class, during the office hours, or by an e-mail *with the course code in the subject line*. In general, phone messages will not receive a response.

Office Hours:

Times of all office hours will be posted on the course web page during the first full week of lectures. Specific times will be announced in-lecture and posted on the course website.

Required Textbook:

-*Calculus: Early Transcendentals, 8th ed.*, by J. Stewart (Brooks/Cole, ISBN 978-1-285-74155-0)

-*Calclabs with Maple for Single Variable Calculus*

(*or Calclabs with Maple Custom Edition, which includes the sections that you will need from both of the above Calclabs manuals.*)

Reference (Optional) Text:

-Student Solutions Manual for the above text(s), both single & multivariable versions, which contains answers to odd-numbered questions

Note: A copy of the textbook and solutions manual will be made available on reserve in Thode Library

Course Software:

Maple 11 (*or any later version. Earlier versions of Maple cannot be used.*)

Maple can be purchased at the campus bookstore or online directly from Maplesoft's website: http://www.maplesoft.com/

Maple is also available to use for free on most of the computer labs on campus. Further information about labs is available on the Lab information page.

Objectives:

To learn the basic concepts and skills associated with sequences, series, polar and parametric equations and basic differential equations.

Topics:

Assorted topics will be covered from chapters 9 through 11, chapters 14 and 15 and selected appendices. See the course calendar on the website for an exact topic list.

Lectures:

There will be one lecture sections of 1AA3, and three sections of 1ZB3. Each will have two, three hour lectures per week:

Course Code	Section #	Lecture Day	Time	Location
1AA3	C01	Tues., Thurs., Fri.	2:30-3:20pm (14:30-15:20)	JHE 376
1ZB3	C01	Tues., Thurs., Fri.	8:30-9:20am (8:30-9:20)	TSH/120
1ZB3	C02	Tues., Thurs., Fri.	11:30am-12:20pm (11:30-12:20)	HSC/1A1
1ZB3	C03	Tues., Thurs., Fri.	2:30-3:20pm (14:30-15:20)	TSH/B128

Tutorials:

There will be one tutorial sections for 1AA3 and three for 1ZB3. Each has one 50 minute session per week:

Course Code	Section #	Lecture Day	Time	Location
1AA3	T01	Tues.	8:30-9:20am (8:30-9:20)	HSC/1A1
1ZB3	T01	Fri.	10:30-11:20am (10:30-11:20)	CNH/104
1ZB3	T02	Wed.	9:30-10:20am (9:30-10:20)	HSC/1A1
1ZB3	T03	Tues.	5:30-6:20pm (17:30-18:20)	~~TSH/B128~~ BSB 147

Math Help Centre:

Weekday afternoons there are TA's available to answer your questions and assist you in the Help Centre in Hamilton Hall. During the summer, The Centre is moved, and now located on the main (second0 foor of Hamilton Hall, in the area near the main entrance behind the Math Cafe.
More information can be found at the Help Centre website:

http://www.math.mcmaster.ca/HelpCentre/

Homework

Assignments:

Approximately every other week a problem set will be assigned. These problems will be both assigned and completed through our online assignment system. Questions will be very similar to the practice problems from the chapter end questions in the course textbook, and the question types that will be appearing on your tests.

Availability, due date, and the exact assignment questions will all be posted on the course website, so please make sure you read the announcements regularly.

These problem sets are MANDITORY and will be used as part of the course grade for ALL students. So do NOT forget to complete an assignment.

Laboratory Assignments:

In addition to the regular online assignments, there will be four (4) Maple "laboratory" assignments which will require the use of Maple(version 11 or later). These will be submitted using the online lab system linked on the course web page. You do not have to attend any scheduled lab times, but TAs will be available if you need help at the times given on the Lab information page.

Due dates are posted in the course calendar, and this and all other information on the Maple assignments and the assignments themselves will be available on the course web page.

Supplemental Problems:

Practice problems for each section of the course are posted on the course website. These materials pertain to the chapters covered. Note that these questions are ***not*** assignments, and thus will neither be asked to be handed in, nor graded. Nevertheless, it is ***very important*** that you at least attempt these problems as they are assigned to assist your familiarization with the course material.

Tests:

There will be **TWO** (2) term tests, each approximately 90 minutes in length. They have **Tentatively** been scheduled to take place on the dates of **Monday, February 8th** , and **Thursday, March 17th** All tests will be held during the evenings outside of class time. These are tentative dates, and as such may be adjusted during the term. Any changes to the dates, as well as the times, locations, and test content *will be announced in lecture* and posted on the course website at least one week before the actual test date.

Calculator Policy:

Use of the "McMaster Standard Calculator", the Casio FX991, any version, will permitted on tests and examinations, unless otherwise stated on our test outlines, ***All other calculators are prohibited.***

Final Examination:

There will be an April Final Examination. This will be a cumulative final exam, two and a half (2.5) hours in duration and covering all materials from the term. Final date, time, and composition will all be confirmed in the weeks leading up to the examinations.

Remember: You are responsible for bringing your student ID to all tests and exams.

Grading Scheme:

Students will be graded according to the following scheme:

5 Online Assignments	5x2%	=	10%
4 Computer Laboratories	4x2.5%	=	10%
2 Term Tests	2x20%	=	40%
Final Examination		=	40%
Total		=	100%

Note: The instructor reserves the right to change the weight of any portion of this marking scheme.

At the end of the course the grades may be adjusted but this will be done uniformly. We will use the grade equivalence chart published in the Undergraduate Calendar to convert between percentages and letter grades.

"Make-up" Work/Assignments:

There are no "make-up assignments " or similar projects available in the course. All contribution to the grades will come only from the tests, assignments, MAPLE 'laboratories' and the examination: no other sources will be considered.

Excused Absences:

If you are absent from the university for a minor medical reason, or non-medical (personal) situations lasting fewer than **3** days, you may report your absence,***once per term***, without documentation, using the McMaster Student Absence Form.

Absences for a longer duration or for other reasons must be reported to your Faculty/Program office, with documentation, and relief from term work may not necessarily be granted.

Please note that the MSAF may not be used for term work worth 25% or more, nor can it be used for the final examination.

The MSAF can be found at: https://pinjap01.mcmaster.ca/msaf/

And further information regarding MSAF policies can be found at: This Web Site

When using the MSAF for this course, please report your absence immediately to Chris McLean, the course coordinator, at mcleac3@math.mcmaster.ca (normally within 2 working days) to confirm the absence.

In the event of such an absence, no make up test will be given, but your course grade will be re-weighted by increasing the weight of the final examination to compensate for the missed test.

Marks and Mark Corrections:

As the term progresses, all test marks will be posted on the course website. It is your responsibility to check for errors in the grades before the day of the final exam, and to report any discrepancies to your instructor. *No*

errors will be corrected unless reported by this time.

In The Event of Unforeseen Circumstances:

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites ***at least weekly*** during the term and to note any changes.

Academic Dishonesty Policy:

Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences. ie: The grade of zero on an assignment, loss of credit with a notation on the transcript (reads " Grade of F assigned for Academic Dishonesty "), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3, located at:

http://www.mcmaster.ca/policy/Students-AcademicStudies/AcademicIntegrity.pdf

-Plagiarism, e.g. the submission of work that is not one's own or for which other credits has been obtained.

-Improper collaboration in group work.

-Copying or using unauthorized aids in tests and examinations.

Math 1AA3/1ZB3 Course Calendar – Winter 2016

(Timing is VERY approximate and WILL be subject to adjustment)

Updated Jan. 1st, 2016

Week	Dates	Topic
Week #1	January 4-8	**Introduction** - Who, What, Where, Why, When? **Lectures:** - 7.5 Integral review - 7.8 Improper Integrals - App. E Induction
Week #2	January 11-15	**Lectures:** - 11.1 Sequences *(Omit Defn. 2)* - 11.2 Series
Week #3	January 18-22	***Approximate Assignment #1 Due Date: Thurs. January 21*** **Lectures:** - 11.2 Series (Continued) - 11.3 Integral Test & Sum Estimates - 11.4 Comparison Tests *(Omit sum estimates)*
Week #4	January 25-29	***Approximate Lab #1 Due Date: Thursday, January 28*** **Lectures:** - 11.5 Alternating series - 11.6 Absolute Convergence, Ratio & Root tests - 11.8 Power Series
Week #5	February 1-5	***Approximate Assignment #2 Due Date: Thursday, Feb. 4*** **Lectures:** - 11.9 Functions as Power Series *(Omit example 8b))* - 11.10 Taylor & MacLaurin *(Omit Mult. and Division of Series))*
Week #6	February 8-12	**Test #1 - Monday, February 8th: (Tentative date)** ***1.5hr (90 min) duration, in the evening. See Announcements for details*** **Lectures:** - 11.11 Taylor Polynomials and Error *(Omit other*

		applications) - 8.2 Surface Area of Revolution - 8.3 Hydrostatic Force and Pressure *(Omit other applications)*
Week #7	February 15-19	***READING WEEK, NO CLASSES***
Week #8	February 22-26	***Approximate Laboratory #2 Due Date: Thursday Feb. 25*** **Lectures:** - 9.1 Intro. to ODE & Modeling - 9.3 Separable ODE - 3.8 Exponential Growth and Decay
Week #9	Feb. 29-March 4	***Approximate Assignment #3 Due Date: Thursday March 3*** **Lectures:** - 9.5 Linear ODE - 10.1 Parametric Equations - 10.2 Calculus of Parametric Curves
Week #10	March 7-11	***Approximate Lab #3 Due Date: Thursday March 10*** **Lectures:** - 10.2 Calculus of Parametric Curves (Continued) - 10.3 Polar Functions - 10.5 Conic Sections - 14.1 Multivariate Functions
Week #11	March 14-18	**Test #2 - Thursday, March 17th: (Tentative date)** ***1.5hr (90 min) duration, in the evening. See Announcements for details*** **Lectures:** - 14.1 Multivariate Functions (Continued) - 2.3 Squeeze Theorem - 14.2 Limits/Continuity in Three Dimensions - 14.3 Partial Derivatives *(Omit the Cobb-Douglas Production Function)*
Week #12	March 21-24	***Approximate Assignment #4 Due Date: Thursday, March 24**** **Lectures:** - 14.4 Tangent Planes and Linear Approx. - 14.5 Multivariate Chain Rule ***Good Friday Holiday, March 25 - No Classes***

Week #13	March 28 - April 1	***Approximate Laboratory #4 Due Date: Thursday, March 31*** **Lectures:** - 14.6 Gradient and $D_u f$ *(Omit Tangent Planes and Gradients)* - 15.1 Multivariate Riemann Sums and Double Integrals
Week #14	April 4-8	***Approximate Assignment #5 Due Date: Thursday, April 7*** **Lectures:** - 15.2 Integrals on General Regions (Type I and II) - Catch up and review

MATH 1B03/MATH 1ZC3 Information Sheet (Winter 2016)

Section				Instructor	Office	Phone	Em
Math1B03 C01	Mo Th 9:30	Tu 10:30	BSB/147	David Lozinski	HH/315	23409	lo
Math1ZC3 C01	Mo Th 9:30	Tu 10:30	CNH/104	Chris McLean	BSB/B124	20142	mc
Math1ZC3 C02	Mo Th 12:30	Tu 13:30	HSC/1A1	Geo Tam	HH/414	26056	ge
Math1ZC3 C03	Mo Th 15:30	Tu 16:30	TSH/B128	Guillaume Blanchet	HH/412	26373	gu

Course Web Page: Available on Avenue to Learn

Textbook:

Elementary Linear Algebra: Applications Version, 11th edition, by Howard Anton and Chris Rorres, published by John Wiley & Sons, Inc. Optionally, the supplement *Student Solutions Manual* with solutions to odd numbered problems is available separately. An online-only version of the textbook can also be rented from CourseSmart.
THe 10th edition will also be supported

Topics:

Vector spaces given by solutions to linear systems. Linear independence, dimension. Determinants. Eigenvalues, eigenvectors and diagonalisation. Complex numbers.

Supporting problems:

Supporting problems will be posted on the course webpage. It is important to attempt as many of the supporting problems as you can. This is essential to the learning process. Answers to some supporting problems can be found in the textbook; a more full solution manual is available separately.

Tutorials:

There will be a large one hour tutorial each week. The tutorials are intended to provide additional material to help students learn the course material, and provide opportunities to ask additional questions and seek help. People ask if the tutorials are mandatory; they are not mandatory, and passing the course is not mandatory either. Attend all lectures and tutorials for the best success.
More personalized assistance can be obtained by coming to the Math Drop-In Centre on the first floor of Hamilton Hall. More detailed times and information are available on their web site:
http://www.math.mcmaster.ca/~mcleac3/Site/HelpCentre.html

Assignments:

There will be 5 assignments made available through online submission. They will be automatically graded if submitted before the deadline expires. A separate link to this is provided on the course web site.

Labs:

In addition to the assignments, there will be 5 labs which will require the use of Matlab (version 7 or later). These will be submitted using the online lab system. You do not have to attend any scheduled lab times. But TAs will be available if you need help at the times given on the Lab information page

Purchasing Matlab:

Matlab can be purchased at the campus bookstore or online directly from Mathworks http://www.mathworks.com/store/default.do. Matlab is also available in most of the computer labs on campus.
All information about labs is available on the Lab information page.

Test:

There will be two tests, tentatively set for the evening of:
Thursday evening, February 25.
Thursday evening, March 24.
The topics covered on the test will be announced on the course webpage.
Calculators are NOT ALLOWED on any of the tests or the exam.

Final Examination:

The final examination (duration 2.5 hours) will be scheduled by the registrar. The registrar will publish more information on the exams at a later date.The exam will cover all the material from the course; details on topics covered will be announced on the course webpage.

Marking scheme:

```
Final examination:   40%
Mid-term tests: 2 at 20% each
Labs:           5 at 2%  each
Assignments:    5 at 2%  each
```

Official Policies:

Exemptions from course work Undergraduate students who are absent from the university for a minor medical reason, lasting fewer than 3 days, may report their absence, once per term, without documentation, using the McMaster Student Absence Form (MSAF). A second absence in a term, or absences for a longer duration or for non-medical reasons must be reported to the Associate Dean of your Faculty/Program office, with documentation, and relief from term work may not necessarily be granted. When using the MSAF, report your absence to your instructor immediately (normally within 2 working days) by email to confirm receipt of the notice. The MSAF can be submitted through the Mosaic system. More information is available at http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work. If potential abuse is detected, the request for relief will be denied and the student will be asked to meet with the Associate or Assistant Dean. For material missed with appropriate documentation, the missing grade will be replaced with the grade on the final exam.

Academic ethics
Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: \Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishon- esty. For information on the various kinds of a academic dishonesty please refer to the Academic Integrity Policy (http://www.mcmaster.ca/senate/academic/ac integrity.htm) speci¯cally Appendix 3. The following illustrates only three forms of academic dishonesty: 1. Plagiarism, e.g., the submission of work that is not one's own or which has been used for other credit. 2. Improper collaboration in group work. 3. Copying or using unauthorized aids in tests and examinations.

Course Modifications
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MATH 1ZC3/1B03 - Rough summary of topics covered
Elementary Linear Algebra - Applications Version - 11th Edition - Anton and Rorres

Matlab

Chapter 1: Systems of Linear Equations and Matrices

- 1.1 Introduction to Systems of Linear Equations - Augmented matrices and elementary operations
- 1.2 Gaussian Elimination - Pivoting, leading ones, infinite solutions, no solutions, unique solutions
- 1.3 Matrices and Matrix Operations - matrix addition, multiplication (order counts!), transpose, trace, linear combination of column vectors
- 1.4 Inverses - Properties of inverse, inverse of product, 2x2 inverse, powers
- 1.5 Elementary Matrices - finding the inverse, factoring matrices
- 1.6 Systems of Equations and Invertibility - Solving systems $A\mathbf{x} = \mathbf{b}$, start of list of equivalencies
- 1.7 Special Matrices - Upper and lower triangular, diagonal, and symmetric matrices

Chapter 2: Determinants

- 2.1 Cofactor Expansion - Cofactor expansion to find determinant
- 2.2 Row Reduction effects on the determinant
- 2.3 Properties of Determinants - Determinant of a product, adjoint, Cramer's rule.

Chapter 5: Eigenvalues and Eigenvectors

- 5.1 Eigenvalues and Eigenvectors - WHAT IS AN EIGENVECTOR? $A\mathbf{x} = \lambda\mathbf{x}$
- 5.2 Diagonalization - PDP^{-1}, matrices raised to powers, diagonalizability
- 5.5 and Online Modules - Dynamical Systems, $\mathbf{v}_{t+1} = A\mathbf{v}_t$, use and role of eigenvectors, $b_1\lambda_1^k\mathbf{x}_1 + ...$, steady state

Appendix B: Complex Numbers Better - 9th Edition Chapter 10, Online modules
The 9th edition is available online on Avenue.

- 9th ed. 10.1 Complex numbers - complex plane, addition, subtraction, multiplication of rectangular complex numbers
- 9th ed. 10.2 More complex numbers - Complex conjugate, modulus of a complex number (absolute value), division by $a + bi$
- 9th ed. 10.3 Polar form - modulus r, argument θ, conversion between forms, non-unique representation, products and quotients, the n nth roots of a number, $e^{i\theta} = \cos\theta + i\sin\theta$

Chapter 3: Vectors in 2-Space and 3-Space

- 3.1 Vectors - sum, difference, components, scalar multiplication, arithmetic
- 3.2 Norm, dot product, and Distance - vector length and distance between points, dot product two ways
- 3.3 Orthogonality - Orthogonal vectors, plane as linear equation in 3-space, normal to a plane, orthogonal projection to find parallel and orthogonal components
- 3.4 Geometry of Linear Systems - lines and planes
- 3.5 Cross product - formula, properties

Chapter 4: General Vector Spaces

- 4.1 Real Vector Spaces - concept, axioms, spaces of R^n, polynomials, functions, matrices
- 4.2 Subspaces - closed under addition and scalar multiplication, linear combinations, the span of a spanning set
- 4.3 Linear Independence - meaning of dependence, definition. The Wronskian will not be on the exam.
- 4.4 Co-ordinates and Basis - minimum spanning set, definition of basis, standard bases, co-ordinates in other bases
- 4.5 Dimension - definition of dimension, build a basis or remove vectors for a basis
- 4.7 Row, Column and Null Spaces - row(A), col(A), using them to find bases, null space and homogeneous solutions, connection between solutions of $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{x} = \mathbf{0}$, consistency
- 4.8 - Rank, nullity, solutions to systems of equations, the list of Equivalent Statements
- 6.3 Gram-Schmidt process and orthogonal projection onto a subspace

Equivalencies for an nxn matrix A (p.299)

- A is invertible (not singular)
- The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$
- A can be carried to I by elementary row operations
- The system $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every choice of $\mathbf{b}$
- A is a product of elementary matrices
- det(A) is not equal to 0
- the columns of A are linearly independent
- the rows of A are linearly independent
- the columns of A are a basis that spans R^n (hence col(A) = R^n)
- the rows of A are a basis that spans R^n
- rank(A) = n
- dim(null(A)) = 0
- no zero eigenvalue

The final exam covers all of the above topics. There are questions from all of the chapters. Note, however, that that some of the earlier sections have fewer explicit questions as that material is used in problems from later sections. There is a higher concentration of questions from chapter 4 (and section 6.3), particularly from those sections taught at the end of the course that have not yet been tested.

Math 1X3 Fall 2014 Course Information

*** This Course Information is also available online at the course web page ***

ms.mcmaster.ca/lovric/1X3.html

You are expected to check the web page often, at least after each lecture
(if you think you will not be able to do it, talk to your lecturer as soon as possible)

Instructor:

Miroslav Lovric (office: Hamilton Hall 411)
email: lovric@mcmaster.ca
office hours will be announced on the course web page

For lecture and tutorial times and locations check the Registrar's website at
https://adweb.cis.mcmaster.ca/mtt/

Lectures AND tutorials are integral parts of the course and you should plan to attend them regularly.

Textbook: "Calculus, Early Transcendentals", 7th edition, by J. Stewart, published by Thomson/Brooks/Cole or Southwestern Publishers. ISBN: 9780538497909

You can use previous editions of the textbook (in particular, 6th edition). In lectures, and for assignments, references to the material in the textbook (page numbers, exercise numbers, etc.) will be based on the new, 7th edition. The same textbook is used for Math 1XX3.

Material covered in the course (selection from the following chapters):

- Elementary functions and models (Chapter 1)
- Limits, continuity and derivatives (Chapters 2, 3)
- Applications of derivatives (Chapter 4)
- Integrals (Chapters 5, 6, 7)
- Material from other chapters as needed

Homework assignments:

Homework assignments are not collected and marked for credit. However, they are an integral part of the course and you should work on them regularly.

- You are allowed to use any calculator or software to work on homework questions
- Practice to write well-organized and readable solutions. In particular, you have to justify main steps in your solution: refer to definitions (do not restate them, just identify), rules and known properties. You will learn how to do this by experience (takes time!).
- Suggested completion dates will be announced on the course web page
- Solutions to all assignments will be posted on the course web page

Quizzes:

- Quiz dates will be announced in class and on the course web page
- Details (e.g., material that will be covered) will be announced on the course web page about a week before a quiz
- Standard McMaster calculator Casio fx 991 may be used during quizzes

Tests:

- For test dates, consult the course web page (in fall 2014 there will be two term tests)
- Details (e.g., material that will be covered, test locations, etc.) will be announced on the course web page about a week before a test
- Tests might be written in the evenings
- You must bring your ID to each test
- Standard McMaster calculator Casio fx 991 may be used during tests

Final Examination:

- Time/day will be scheduled by the Registrar and announced on the course web page as soon as the information becomes available.
- Details (e.g., material that will be covered, final examination locations, etc.) will be given in class and announced on the course web page
- Standard McMaster calculator, Casio fx 991MS+ may be used during final exam

Course Evaluation:

Quizzes	25%
Tests (15% each)	30%
Final Exam	45%

The instructor reserves the right to change the weight of any portion of this marking scheme. For students in good academic standing, other weights might be considered. In either case, the final mark will be computed using this weighting and the new weighting(s). The highest score for a particular student will be her/his final mark.

At the end of the course the grades may be adjusted but this can only increase your grade and will be done uniformly. We will use the grade equivalence chart published in the Undergraduate Calendar to convert between percentages and letter grades.

In case of difficulty/problems:

Contact your instructor or your teaching assistant as soon as possible. Failing that, contact the Associate Dean's Office in BSB-133.

Policy regarding missed work:

If you have missed work, it is your responsibility to take action.

If you are absent from the university for a minor medical reason, lasting fewer than 5 days, you may report your absence, once per term, without documentation, using the McMaster Student Absence Form. Absences for a longer duration or for other reasons must be reported to your Faculty/Program office, with documentation, and relief from

term work may not necessarily be granted. When using the MSAF, report your absence to lovric@mcmaster.ca. You must then contact
Dr. Lovric immediately (normally within 2 working days) by email at lovric@mcmaster.ca to learn what relief may be granted for the work you have missed. Details will be posted on the course web page. Please note that the MSAF may not be used for term work worth 30% or more, nor can it be used for the final examination.

Please note! Once a final examination is written, the final grade cannot be adjusted to take into account any special situation.

Academic dishonesty:

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at http://www.mcmaster.ca/academicintegrity/

The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Your marks:

At the end of the term, all grades in the course will be posted (by student number). It is <u>your responsibility</u> to check for errors <u>before the day of the final exam,</u> and to report any discrepancies to your instructor or to your TA. <u>No error will be corrected</u> unless reported by this time.

Math 1XX03 / Calculus for Math and Stats II

Prof. S. Alama / HH-421 / x27363 / alama@mcmaster.ca

Website: http://www.math.mcmaster.ca/alamas/main/m1xx3/m1xx3.html

Office hours TBA

Lectures: Tues, Thurs, Fri 11:30-12:20 in HH/302

The content of the lectures is the content of the course. You are expected to attend all lectures, and it is your responsibility to obtain lecture notes from a classmate in case of absence. A table recording each lecture's topic may be found on the course web page under the link "Where are we now?"

Course Materials:

- Calculus: Early transcendentals, 7th edition, by James Stewart, Thomson Brooks/Cole.

The course will cover material selected from chapters 5-14, including: the Definite Integral and Fundamental Theorem of Calculus, Series, Partial Derivatives, and (time permitting) Parametric Equations.

- WebAssign. *We cannot require you to purchase it, but students are* ***very strongly encouraged*** *to subscribe to WebAssign. WebAssign is a system for on-line assignments which provides many advantages for students, including instant feedback, extra practice problems, and hyperreferences to the appropriate sections in the text. An access code for WebAssign may be purchased at the McMaster Campus Store or online directly at* http://www.webassign.net *Our Class Key is:* **mcmaster 8294 9902**

Objectives: To learn and apply the ideas and techniques of differential and integral calculus to solve problems. Some stress will be placed on the underlying ideas and theoretical framework of the subject to prepare students for more advanced courses in mathematics and statistics.

Tutorials: *Tutorials begin on Monday January 12, 2015.*

T01=Mon 09:30-10:20, HH/102
T02=Thurs 15:30-16:20, T13/106
T03=Thurs 16:30-17:20, HH/102

The tutorials are intended to provide additional examples to supplement the course material, and to provide additional opportunities for students to ask additional questions and seek help. Although attendance in tutorials is not mandatory, it is strongly encouraged.

Help? More personalized assistance can be obtained by coming to the **Math Drop-In Centre** on the first floor of Hamilton Hall. The hours will be:

Monday-Thursday: 2:30 pm - 8:30 pm, Friday: 2:30 pm - 6:30 pm

Assignments and Practice Problems: There will be periodic homework assignments, announced in class and on the website, and all students should do the problems in order to prepare for the tests and exam. Students who opt to purchase WebAssign will complete their assignments online; problems are corrected immediately and students have several attempts to correctly answer each problem. Equivalent problems drawn from the Stewart text will be posted on the course website for those students who opt out of WebAssign; note that **these will not be collected and will not be marked**. Students who opt out of

WebAssign are encouraged to make good use of the tutorials, Office Hours, and Math Help Centre hours in order to obtain feedback on their preparation. Additional practice problems will be also be given, both via WebAssign and from the textbook and announced on the course webpage. The practice problems will not be collected, but you will be responsible for their content on the tests and final exam.

Marking Scheme:

- ✓Three midterm tests 3 x 16.67% = 50%
- ✓Final Examination 40%
- ✓Maximum among {Midterm grades, Final Exam grade, WebAssign average}=10%

For those who do not subscribe to WebAssign, the final grade will be computed as above with no WebAssign score.

Alternative marking schemes may also be used, in which case your final grade will be given by the maximum mark obtained among all schemes considered.

Midterm Tests: There are 3 midterm tests, tentatively scheduled for the normal class time on

Thursday January 29, Thursday March 5, and Friday March 27

The topics covered will be announced in class and on the course web page. You MUST bring your student ID to all tests, or your mark will be zero.

Final Exam: A 3-hour final exam administered by the registrar will cover all course material. The date, time, and location will be announced by the registrar's office sometime later in the term.

Calculators: Only the standard Casio fx 991 calculator is permitted during midterm tests and the final exam. (You may use any calculating device your heart desires for homework and studying.)

Excused Absences: Students seeking an exemption for an assignment, project or mid-term test must contact the office of the Associate Dean (Studies) of their Faculty within a week after the due or test date with proper documentation, e.g., a doctor's note. I will act on the exemption only after receiving official communication from the relevant Associate Dean's office that the grounds for exemption have been accepted. Deferrals for the final exam are handled exclusively by the Associate Dean's office. The percentages for exempted tests will be transferred to the final exam.

The following information concerns the on-line MSAF absence reporting tool:

For absences from classes lasting up to 5 days: Using the McMaster student absence form **(MSAF)** on-line, self-reporting tool, **undergraduate students** may report absences lasting up to 5 days and may also request relief for missed academic work. The submission of medical or other types of supporting documentation is normally not required. Students may use this tool to submit a maximum of one request for relief of missed academic work per term. Students must immediately follow up with their course instructors regarding the nature of the relief. Failure to do so may negate the opportunity for relief. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work in his/her course.

For absences from classes lasting more than five days:

Students who are absent more than five days cannot use the on-line, self-reporting tool to request relief. They MUST report to their Faculty Office to discuss their situation and may be required to provide appropriate supporting documentation. If warranted, students will be approved to use a discretionary version of the MSAF on-line, self-reporting tool.

Ethics: Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of a academic dishonesty please refer to the Academic Integrity Policy:

http://www.mcmaster.ca/univsec/policy/AcademicIntegrity.pdf
specifically Appendix 3. The following illustrates only three forms of academic dishonesty:
1. Plagiarism, e.g., the submission of work that is not one's own or which has been used for other credit.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Disclaimer: The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Mathematics 2FM3 Introduction to Mathematical Finance Fall 2014

Instructor: Dr. Petar Jevtic (pjevtic@math.mcmaster.ca) HH 416
Office Hours: 10:30-11:30 We and Th.

Lectures will take place in HH 302 starting Friday September 4th at the following times: Mo, We and Th 13:30-14:20.

Textbook: Mathematics of Investment and Credit, 5th Edition by Samuel A. Broverman, published by ACTEX Publication Inc., Winsted CT. ISBN:978-1-56698-767-7.

Topics: Nominal and effective rates of interest and discount, forces of interest and discount, compound interest, annuities certain; amortization, sinking funds, bonds, security evaluation, determination of yields.

Tutorials: Attendance at one tutorial per week is mandatory, starting in the week of September 16th These take place at the following times/rooms:
T01 Fr 08:30-09:20 HH/305
T02 Tu 10:30-11:20 HH/305
T03 Tu 13:30-14:20 HH/305

Assignments/Homework, Midterm Tests and Final Exam:
There are no assignments or homework to be submitted. Practice problem sets for each week will be posted on the course website. They will pertain to the chapters covered. Note that these problems sets are not assignments, and will not be handed in or graded. Nevertheless, it is very important that you practice these problems as they are designed to assist you with the course material. Most of problems you will find on the tests are selected from the practice problem sets.

There will be two tests during the term. The composition of the tests will be announced on the course website and in lecture at a later date.

Test #1 (*Tentative*):	Test #2 (*Tentative*):
Date: Wednesday, October 15, 2014	Date: Thursday, November 13, 2014
Test Time: 13:30-14:20	Test Time: 13:30-14:20

The Final Exam will last three hours and will cover all material from the term. It will be scheduled by the registrar. Location, final date, time, and composition will all be announced both on the course website and in lecture at a later date.
Don't forget: You are responsible for bringing your student ID to all tests and exams.

Final Marking Scheme:
Your final grade for the course will be determined by the following scheme:
Tests: 20% each, Final Examination: 60%

Calculator Policy: Use of the McMaster Standard Calculator the Casio FX991, is permitted on tests and examinations. All other calculators are prohibited.

Excused Absences: Exemption from a test or an assignment for valid reasons is possible. In the event of an exemption, NO make-up test or assignment will be given. Instead

your course grade will be reweighed by increasing the weight of the final examination to compensate for the missed material. It is strongly recommended that your take all tests for your best grade.

Requests for Relief for Missed Academic Term Work: If you are absent from the university for a minor medical reason, lasting fewer than 5 days, you may report your absence, once per term, without documentation, using the McMaster Student Absence Form (MSAF). When using the MSAF, report your absence immediately (e.g. by e-mail within 2 days) follow up with their course instructors regarding the nature of the relief. Failure to do so may negate the opportunity for relief. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work in his/her course.

Students who are absent more than five days cannot use the on-line, self-reporting tool to request relief. They MUST report to their Faculty Office to discuss their situation and may be required to provide appropriate supporting documentation. Please note that the MSAF may not be used for term work worth 30% or more, nor can it be used for the final examination. The relevant link is `https://www.mcmaster.ca/msaf/index.html`.

Academic ethics: Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of a academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3:
`www.mcmaster.ca/policy/Students-AcademicStudies/AcademicIntegrity.pdf`
The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g., the submission of work that is not one's own or for which other credit has been obtained.

2. Improper collaboration in group work.

3. Copying or using unauthorized aids in tests and examinations.

4. Collaboration in written assignments - Discussions about assignments are allowed and are generally beneficial. However, you must write up solutions of the assignment problems by yourself and in your own words. Copying with minor changes (e.g., with symbols changed or with slightly different wording) from solutions prepared by another person or publication, in whatever format, will be dealt with as an act of plagiarism.

Important Message:
The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Math 2T3 **Numerical Analysis I** Winter 2016

Dr. N. Kevlahan

Lectures: Mondays 12:30-13:20, Tuesdays 13:30-14:20 and Thursdays 12:30-13:20 in Hamilton Hall 305
Tutorial/Lab: Fridays 14:30-15:20 in Hamilton Hall 305

Office: HH 324, Tel: x23412
Email: `kevlahan@mcmaster.ca`
Course website: `Avenue to Learn`
Office hours: by appointment

Purpose of the course

Most problems in physics, engineering and applied mathematics are formulated in terms of equations (e.g. ordinary differential, partial differential, or stochastic) which cannot be solved analytically. However, one can almost always reduce this one difficult problem to many simple problems which can be solved numerically on a computer. More precisely, we reduce the problem to a large system of linear algebraic equations.

For an excellent explanation of what numerical analysis *really* is read Trefethen's The Definition of Numerical analysis. For a nice introduction to Numerical Analysis, I suggest reading Trefethen's entry in the *Princeton Companion to Mathematics*.

This course will teach you efficient methods for solving very large systems of linear algebraic equations. With this knowledge as a foundation you will be able to easily tackle any advanced course in scientific computation or numerical analysis. You will also learn the basic mathematical concepts of numerical error analysis (i.e. sensitivity, conditioning and accuracy) and estimation of computational complexity, which are essential for developing a useful numerical algorithm.

The course material will be a mix of applied problems, theoretical analysis, algorithm development, and practical programming. Matlab will be the computer language used for all examples and assignments, as it has been developed explicitly for numerical linear algebra.

Text

The primary text for the course is the **Math 2T03 Courseware**, available from the Bookstore.

(The courseware is an updated 2nd edition selection from: M.R. Grasselli & D.E. Pelinovsky (2008) *Numerical mathematics*, Jones & Bartlett.)

The supplementary texts for the course are:
M.T. Heath (2002) *Scientific computing: an introductory survey*, second edition, McGraw-Hill.

R. Pratap (2002) *Getting started with matlab: a quick introduction for scientists and engineers*, Oxford University Press.

A guide to Matlab (e-book)

Software

Student edition of Matlab. We highly recommend that you buy this software. `matlab` is an essential part of this course, and will be very useful in any future mathematically intensive courses. Of course, you will also be able to use `matlab` in the University's computer labs.

If you don't have `matlab` on your computer and don't want to buy the student version you can also use `Octave` https://www.gnu.org/software/octave/, which is very similar.

Outline

The course is organized as follows (note that timings and content are tentative). Sections refer to the courseware.

1. **Introduction to scientific computation using Matlab** (`m-file` tutorials from website, A guide to Matlab, §1.1-1.4, 4 lectures)
   - The `matlab` session.
   - Using `matlab` help (e.g. built-in functions).
   - Working with matrices and vectors.
   - Advanced data structures: structure arrays, field arrays, cells.
   - Creating and printing figures.
   - Basic control structures (sequential, repetition, selection).
   - Editing, saving and executing scripts.
   - Functions, sub functions and cell mode.
   - Some useful built-in functions.
2. **Numerical solution of linear systems of equations** (Chapters 1 and 2, 10 lectures)
   - Floating pointing point arithmetic. (§1.5)
   - Finite dimensional vectors spaces, linear maps, vector and matrix norms, sensitivity and conditioning, error bounds (§2.1-2.5).
   - Direct methods: LU factorization (§2.2), computational complexity, sensitivity (§2.5).
   - Iterative methods for solving linear systems: Jacobi, Gauss-Seidel, SOR (§2.4).
3. **Nonlinear equations (zeros of a function)** (Chapter 5, 3 lectures)
   - One dimension: interval bisection, fixed-point iteration, Newton's method (§5.1, 5.2).
   - Multi-dimensions (§5.3).
4. **Matrix factorizations, orthogonality and data fitting by least squares** (Chapter 3, 10 lectures)
   - Least-squares method, data fitting (§3.5).
   - Symmetric and orthogonal matrices (§3.1).
   - Orthogonal projections (§3.2).
   - QR factorization (§3.3, 3.4).
   - Cholesky factorization (§3.6).
5. **Eigenvalues and eigenvectors** (Chapter 4, 10 lectures)
   - Basic properties (§4.1–4.3).
   - Power iteration, Rayleigh quotient iteration, deflation (§4.4).
   - Simultaneous iteration, orthogonal iteration, QR iteration, Hessenberg reduction (§4.5).
   - Singular Value Decomposition (§4.6).
   - Sensitivity (§4.7).

Evaluation

There will be six bi-weekly assignments, one mid-term test, and a final exam.

Assignments

Six problem sheets will be given and marked for credit. Each assignment will have a significant Matlab component. Assignments are to be dropped into the Math 2T3 locker in the basement of the Hamilton Hall by 15:00 on the due date. No late assignments will be accepted. Solutions to assignments and the test will be posted on the course webpage. The tentative assignment schedule is as follows:

Assignment posted	**Assignment due**
January 7	January 21
January 21	February 4
February 4	February 22
February 25	March 10
March 10	March 24
March 24	April 7

Test

There will be one 50 minute test (during the regular class time):

Thursday 25 February, 12:30-13:20 in T29 101 (tentative)

Final exam

There will be a 2.5-hour final examination during the April examination period. Only standard McMaster University calculators (Casio FX-991) may be used in the final examination.

Grading system

The final mark will be calculated as follows:

Homework	35%
Test	15%
Final exam	50%

I reserve the right to change the weight of any portion of this marking scheme. If changes are made, your grade will be calculated using the original weightings and the new weightings, and you will be given the higher of the two grades. At the end of the course the grades may be adjusted but this can only increase your grade and will be done uniformly. I will use the grade equivalence chart in the university calendar to convert between letter grades, grade points and percentages.

Important message

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Official notices

Academic integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

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2. Improper collaboration in group work.

3. Copying or using unauthorized aids in tests and examinations.

Academic accommodation of students with disabilities language for course outlines

Students who require academic accommodation must contact Student Accessibility Services (SAS) to make arrangements with a Program Coordinator. Academic accommodations must be arranged for each term of study. Student Accessibility Services can be contacted by phone 905-525-9140 ext. 28652 or e-mail sas@mcmaster.ca. For further information, consult McMaster University's Policy for Academic Accommodation of Students with Disabilities.

Absences and missed work (MSAF)

In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work". Please note these regulations have changed beginning Fall 2015.

Please follow this link for more information:

McMaster University
Math 2X3 (Term I, 2015-2016)
Course Information Sheet

Instructor: M. Wang
Office: HH 321
Extension: 23405
e-mail: wang@mcmaster.ca

Office Hours: M 14:30 - 16:00 ; Th 14:30 – 15:20 or by appointment

Lectures: MW 8:30 – 9:20; F 10:30-11:20; in JHE 264

Tutorials:

Tut 1 T 12:30 – 13:20	HH 305	Sean Conley
Tut 2 M 16:30 -- 17:20	HH 305	Vincent Chiu
Tut 3 W 16:30 – 17:20	HH 305	Vincent Chiu
Tut 4 Th 9:30 – 10:20	HH 305	Sean Conley

Course Webpage:

`http://ms.mcmaster.ca/mckenzie/Math2X3-2015.htm`

Required Text:

J. Stewart, *Calculus: Early Transcendentals,* (7th edition), Brooks/Cole

The 8th edition is perfectly fine to learn out of, but assignment and practice problems will refer to the 7th edition. There is also an optional student study guide which may be helpful. A copy of the 7th edition has been put on reserve at the Thode Library.

Objectives and Topics to be Covered:

In this course we will first study multiple integrals and their applications, and then use them to define integrals of vector fields (over curves and surfaces) which are independent of parametrization. These vector integrals are used in turn to study the generalizations of the fundamental theorem of calculus to higher dimensions: the theorems of Green, Stokes and Gauss. The theory will be illustrated by applications.

Assignments:

There will be 5 written assignments whose due dates will be announced in class and posted on the course web-page. Assignments are to be handed in before noon on the due date in the appropriate box in the basement of Hamilton Hall. Assignments deposited after 12:30 pm into the 2X3 box **will not be collected or marked**.

Tests and Exam:

There will be two **tests** held **during lecture periods**. The tentative dates are

Test I : **October 9, 2015** (Friday) from 10:30 – 11:20 (locationTBA)
Test II: **November 13, 2015** (Friday) from 10:30 -- 11:20 (locationTBA)

The topics covered in the two tests will be announced in class and posted on the course web-page.

Students who need to take the tests at the *Student Accessibility Services* must identify themselves to the instructor by September 18, 2015.

A **written final exam** scheduled by the Registrar's office will be held during exam period in December. Only the **Casio fx 991 calculator** may be used during the tests and the final exam. Students must bring their **McMaster ID** cards to the tests and final exam for inspection.

Grading Scheme:

Your **final mark** will be the **higher** of the following two computations:

(1) 5 assignments (15%), 2 tests (36%), final exam (49%),
(2) final exam (100%).

Your final mark will then be converted to a letter grade using a scheme similar in spirit to that published in the Undergraduate Calendar. See section (a) below for the adjusted grading scheme in cases of medical and other exemptions.

Important Message:The instructor and University reserve the right to modify elements of the course during the term. The University may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on the changes. It is the responsibility of all students to check their McMaster email accounts and course webpages weekly during the term and to note any changes.

Policy re Homework and Tests:

(a) MSAF Policy

Occasionally, students have to miss classes, assignments, or tests because of medical or personal non-medical reasons. Students should review and follow the Academic Regulations in the Undergraduate Calendar under "Requests for Relief for Missed Academic Term Work". The link is
http://academiccalendars.romaster.ca/content.php?catoid=13&navoid=2208#Requests for Relief for Missed Academic Term Work

Requests for Relief for Missed Academic Term Work or for absences from classes lasting up to 3 days due to minor medical or personal reasons: Using the McMaster Student Absence Form (MSAF) on-line, self-reporting tool, undergraduate students may report absences lasting up to 3 days and may also request relief for missed academic work. The submission of supporting documentation is not required. Students may use this tool to submit a maximum of one request for relief of missed academic work per term. Students must immediately follow up with their course instructors regarding the nature of the relief. Failure to do so may negate the opportunity for relief. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work in his/her course.

For medical or personal situations **lasting more than three calendar days,** and/or for missed academic work worth 25% or more of the final grade, and/or for any request for relief in a term where the MSAF has been used previously in that term: Students must report to their Faculty Office to discuss their situation and will be required to provide appropriate supportingdocumentation. If warranted, the Faculty Office will approve the absence, and the instructor will determine the appropriate relief.

For Math 2X3, **the composition of computation (1) for your final mark will then be changed as follows. The percentage of an excused test or written assignment will be transferred and added to that of the final exam.**

(b) Academic Ethics

Your attention is drawn to the documents "Senate Statement on Academic Ethics" and "Senate Resolutions on Academic Dishonesty" which you have received during registration and which can be obtained from the Senate Office. Infringements on the rules and principles in these documents will be dealt with in the manner stated therein.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For official information on what constitutes academic dishonesty please refer to the written copy of the document Academic Integrity Policy, an electronic copy of which can be found at the url http://www.mcmaster.ca/univsec/policy/AcademicIntegrity.pdf. The following illustrates only three forms of academic dishonesty:

(i) Plagiarism, e.g., the submission of work that is not one's own or for which other credit has been obtained.
(ii) Improper collaboration in group work.
(iii) Copying or using unauthorized aids in tests and examinations.

(c) Collaboration in Homework Assignments

Discussions about homework assignments are allowed and are generally beneficial. However, you must write up the solutions of the assignment problems **by yourself and in your own words.** Copying with minor changes (e.g. with symbols changed, or with slightly different wording) from solutions prepared by another person, publication, or website, in whatever format, will be dealt with as an act of plagiarism.

Rough Course Schedule

Week	**Topic**	**Sections**
Sept 8	double integrals I	15.1-15.3
Sept 14	double integrals II	15.4-15.6
Sept 21	triple integrals	15.7-15.8
Sept 28	change of variables	15.9-15.10
Oct 5	vector fields and line integrals (Test 1:Oct 9)	16.1-16.2
Oct 19	Line integrals	16.2-16.3
Oct 26	Green's Theorem	16.4
Nov 2	derivatives of vector fields, parametrized surfaces	16.5-16.6
Nov 9	surface integrals (Test 2: Nov 13)	16.7
Nov 16	Stokes' s Theorem	16.8
Nov 23	Stokes's theorem and Gauss's theorem	16.8-16-9
Nov 30	Gauss's Theorem	16.9
Dec 8	Review	

Mathematics 3FM3 **Mathematics of Finance** **Fall 2014**

Instructor: Dr. Petar Jevtic (`pjevtic@math.mcmaster.ca`) HH 416
Office Hours: 10:30-11:30 Tu and Fr.

Lectures will take place in HH 302 starting Friday September 4th at the following times: Tu, We and Fr 9:30-10:20.

Textbook: Stochastic Calculus for Finance I: The Binomial Asset Pricing Model by Steven Shreve, published by Springer 2004 edition (June 28, 2005) ISBN-10: 0387249680.

Topics: Introduction to finance in discrete time: efficient markets and the no arbitrage condition, binomial asset pricing model, conditional expectation, martingales and options.

Assignments/Homework, Midterm Tests and Final Exam:
There will be up to six assignments to be submitted. They will be posted on the course website at the appropriate times. It will be written when they are due. They will pertain to the chapters covered.

There will be one test during the term. The composition of the test will be announced on the course website and in lecture at a later date.

Test #1 (*Tentative*):
Date: Tuesday, November 4th, 2014
Test Time: 9:20-10:20

The Final Exam will last three hours and will cover all material from the term. It will be scheduled by the registrar. Location, final date, time, and composition will all be announced both on the course website and in lecture at a later date.
Don't forget: You are responsible for bringing your student ID to all tests and exams.

Final Marking Scheme:
Your final grade for the course will be determined by the following scheme:
Assigments (in total): 20%, Test: 20%, Final Examination: 60%

Calculator Policy: Use of the McMaster Standard Calculator the Casio FX991, is permitted on tests and examinations. All other calculators are prohibited.

Excused Absences: Exemption from a test or an assignment for valid reasons is possible. In the event of an exemption, NO make-up test or assignment will be given. Instead your course grade will be reweighed by increasing the weight of the final examination to compensate for the missed material. It is strongly recommended that your take all tests for your best grade.

Requests for Relief for Missed Academic Term Work: If you are absent from the university for a minor medical reason, lasting fewer than 5 days, you may report your absence, once per term, without documentation, using the McMaster Student Absence Form (MSAF). When using the MSAF, report your absence immediately (e.g. by e-mail within 2 days) follow up with their course instructors regarding the nature of the relief. Failure to do so may negate the opportunity for relief. It is the prerogative of

the instructor of the course to determine the appropriate relief for missed term work in his/her course.

Students who are absent more than five days cannot use the on-line, self-reporting tool to request relief. They MUST report to their Faculty Office to discuss their situation and may be required to provide appropriate supporting documentation. Please note that the MSAF may not be used for term work worth 30% or more, nor can it be used for the final examination. The relevant link is `https://www.mcmaster.ca/msaf/index.html`.

Academic ethics: Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of a academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3:
`www.mcmaster.ca/policy/Students-AcademicStudies/AcademicIntegrity.pdf`
The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g., the submission of work that is not one's own or for which other credit has been obtained.

2. Improper collaboration in group work.

3. Copying or using unauthorized aids in tests and examinations.

4. Collaboration in written assignments - Discussions about assignments are allowed and are generally beneficial. However, you must write up solutions of the assignment problems by yourself and in your own words. Copying with minor changes (e.g., with symbols changed or with slightly different wording) from solutions prepared by another person or publication, in whatever format, will be dealt with as an act of plagiarism.

Important Message:
The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Math4FM3: Financial Markets and Derivatives
Course Outline - 2014-2015 Term 2

Classes are Tues, Thurs, Fri at 8:30am in HH 217 with D. Lozinski

The course web page can be found on Avenue to Learn

Instructor:
Dr. David Lozinski HH/315, x23409, lozinski@math.mcmaster.ca
Office hours will be after class on Tuesday (9:30am) and Thursday (9:30am), or by appointment.
(There are additional office hours nominally for linear algebra on Tuesdays at 2:30pm and Thursdays at 1:30pm.)

Textbook:
Options, Futures, and other Derivatives, 8th or 9th edition, by John C. Hull, published by Prentice Hall, ISBN 978-0-13-216494-8.

Topics:
Modelling of options, futures, interest rate securities and other financial derivatives in continuous time using Brownian motion and stochastic calculus. Topics include risk-neutral pricing, the Black-Scholes framework, dynamic hedging, volatility and risk.

Mid-Term Test:
There is 1 midterm test, worth 25% of the final grade, tentatively set for Wednesday evening, Feb 25.
More information, including the topics covered, will be announced in class.
You MUST bring your student ID to all tests, or your mark will be zero.

Assignments:
There will be up to 5 assignments to be completed during the term. Collectively they will be worth 5% of the final grade.

Small Project:
Students will be assigned to small groups to research a topic related to the course. A presentation and a write-up will be required, and will be graded for 20% of the final mark. More information about the projects will be provided in class.

Final Exam:
A 3 hour final exam will be scheduled and administered by the registrar in April. It will cover all course material.

Marking Scheme:

Midterm test	25%
Final Examination	47%
Assignments	5%
Class Participation	3%
Small Project	20%

Calculator: Only the standard Casio fx 991 calculator is permitted.

Official Policies:
Requesting Relief for Missed Academic Term Work -
Undergraduate students who are absent from the university for a minor medical reason, lasting fewer than 5 days, may report their absence, once per term, without documentation, using the McMaster Student Absence Form (MSAF). A second absence in a term, or absences for a longer duration or for non-medical reasons must be reported to the Associate Dean of your Faculty/Program office, with documentation, and relief from term work may not necessarily be granted. When using the MSAF, report your absence to Dr. Lozinski immediately (normally within 2 working days) by email at lozinski@math.mcmaster.ca to confirm receipt of the notice. The MSAF can be reached at https://pinjap01.mcmaster.ca/msaf/. If potential abuse is detected, the request for relief will be denied and the student will be asked to meet with the Associate or Assistant Dean. For material missed with appropriate documentation, the missing grade will be replaced with the grade on the final exam.

McMaster Policy on Academic Integrity: You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: Grade of F assigned for academic dishonesty), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty, please refer to the Academic Integrity Policy, located at: www.mcmaster.ca/academicintegrity The following illustrates only three forms of academic dishonesty:
1) Plagiarism e.g., the submission of work that is not one's own or for which other credit has been obtained.
2) Improper collaboration in group work.
3) Copying or using unauthorized aids in tests and examinations.

Course modifications: The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of students to check their McMaster email and course websites weekly during the term and to note any changes.

Mathematics 4MB3/6MB3
Mathematical Biology
Course Information Sheet, Winter 2016

Instructor: David Earn
Office: Hamilton Hall 317
Phone: (905) 525-9140, x27245
E-mail: `earn@math.mcmaster.ca`
Home page: `http://www.math.mcmaster.ca/earn`

TA/Marker: Karsten Hempel
Office: Hamilton Hall 403
E-mail: `hempelkr@math.mcmaster.ca`

Class Location: Hamilton Hall 102

Class Times:

- M W Th: 1:30–2:20pm
- Th 7:00–10:00pm (to be used for midterm test and potentially instead of normal lecture times some weeks; location BSB/B139)

Prerequisites: MATH 3F03 "Advanced Differential Equations" or an equivalent course in the qualitative theory of nonlinear ordinary differential equations.

Course Content: Introduction to mathematical modelling of infectious disease (ID) transmission. Application of ID models to understanding historical epidemics and current problems in infectious disease control. The primary focus will be on deterministic models, but stochastic models will also be discussed. Introduction to software for mathematical typesetting, graphics, simulations, phase portraits and bifurcation diagrams.

Course Objectives:

- To learn to create and analyze mathematical models of biological systems and to relate these models to data from real biological systems.
- To become familiar with some primary research literature in mathematical epidemiology.
- To develop skills and experience in conducting collaborative research in mathematical biology.
- To learn to present the results of mathematical modelling in documents that are prepared using typesetting and graphics software that are standard in the professional mathematics community.

Course wiki: `http://www.math.mcmaster.ca/earn/4MB3`
Course information, including announcements, handouts, lecture slides, assignments, solutions, links to downloadable course-related software, *etc.*, will be available on the course wiki. You are expected check it regularly.

Groups: An important aspect of the course will be to learn to work effectively in small groups (ideally 4 students per group). Groups will be formed early in the course and you will work together on the assignments and final project. Formation of groups will be discussed in class. Individuals will submit a group contribution form after each group assignment and the final project.

Assignments: There will be several assignments (probably 5). Assignments must be submitted on time at the start of the class on the due date. Late assignments **will NOT** be accepted.

Assignment	**Tentative Due Date**
1	Thursday 28 January 2016
2	Thursday 11 February 2016
3	Monday 29 February 2016
4	Monday 14 March 2016

Each group will submit one joint document for each assignment. The document must be typeset in LaTeX and all graphics must be prepared using `R` or `XPPAUT`. Assignments must be submitted both as hardcopy and by e-mail (one e-mail message from each

group, with attachments including all source documents and the final compiled pdf file).

Solutions to selected problems will be distributed by e-mail after the due date. ***Note:*** *Only a selection of problems on each assignment will be marked; your grade on each assignment will be based only on the problems selected for marking. Problems to be marked will be selected after the due date.*

Tests: There will be **one** Term Test:

Tentative Date	**Tentative Time**	**Tentative Location**
Thursday 10 March 2016	7:00–9:00pm	BSB/B139

There will be no make up test. See the policy on excused absences in note 1) below.

Final Project: The most important component of the course is the final project, which will be done in the same groups as the assignments. In addition to the final group project document, each individual will submit her/his own "research notebook" or "lab book" in which s/he has kept track of all work done on the project over the course of the term. The individual notebooks will be due to be submitted to the instructor several times during the term. Details about the project will be posted on the course wiki several weeks into the term.

Final Presentation: Near the end of the term, each group will summarize their project in an oral presentation in class (using slides prepared with the `beamer` package in LaTeX).

Software: In order to complete the assignments and final project, you will be required to develop basic competence with software for mathematical typesetting (LaTeX), graphics and numerical analysis (`R`), and numerical solution of differential equations and bifurcation analysis (`XPPAUT`). These applications are all open-source free software projects and can be downloaded and installed on any computer.

- LaTeX: `http://www.latex-project.org/`
- `R`: `http://www.r-project.org`
- `XPPAUT`: `http://www.math.pitt.edu/~bard/xpp/xpp.html`

You will need to install these applications on your laptop. If you do not have a laptop, let the instructor know immediately.

Course style: During approximately the first half of the term, there will be lectures, some of which will consist of demonstrations/tutorials about the required software. Later in the term many classes will be devoted to group project sessions, i.e., class time set aside for group project work with the instuctor present and available to answer questions.

Communicating with the instructor: You will need to send e-mail messages to the instructor. Bear in mind that the instructor typically receives 100 e-mail messages per day and it is easy for messages to be missed or get backlogged. Every e-mail message you send to the instructor must have a helpful, descriptive subject line. The subject line should always have the form "`Math 4MB3: ...`". Examples might be:

```
Math 4MB3: confusion about assignment 1, problem 2a
Math 4MB3: progress on extra credit problem
Math 4MB3: dog ate our group's project
```

Final Grade: Your final grade will be determined as follows:

Component	**Weight**
Assignments	25%
Term Test	20%
Final Project	40%
Oral Presentation	10%
Participation	5%

Note that participation includes completing online surveys and peer evaluations as required.

Reference list

There is no course textbook. However, the following articles should be helpful:

- **Earn, D.J.D.**, 2004. "Mathematical modelling of recurrent epidemics." *Pi in the Sky*, **8**, 14–17 (Intended audience: Keen high school mathematics students)

- **Earn, D.J.D.**, 2008. "A light introduction to modelling recurrent epidemics." In *Mathematical Epidemiology*, F. Brauer, P. van den Driessche, J. Wu (editors) *Lecture Notes in Mathematics* **1945**, Springer, pp. 3–18 (Intended audience: undergraduate mathematics students)

- **Earn, D.J.D.**, 2009. "Mathematical epidemiology of infectious diseases." In *Mathematical Biology*, M.A. Lewis, M.A.J. Chaplain, J.P. Keener, P.K. Maini (editors) *IAS/Park City Mathematics Series* Volume **14**, American Mathematical Society, pp. 151–186 (Intended audience: senior undergraduate and beginning graduate mathematics students)

These articles and other papers that will be discussed during the course are available at `http://www.math.mcmaster.ca/earn/publications.php`.

The following books may also be useful references:

- "Infectious Diseases of Humans: Dynamics and Control" by Roy Anderson and Robert May (Oxford, 1991).
- "Mathematical models in population biology and epidemiology" by Fred Brauer and Carlos Castillo-Chavez (Springer, 2001).
- "Modeling Infectious Diseases in Humans and Animals" by Matt Keeling and Pej Rohani (Princeton, 2008).
- "Nonlinear Dynamics and Chaos" by Steven H. Strogatz (1994).
- "Simulating, Analyzing, and Animating Dynamical Systems: A Guide to XPPAUT for Researchers and Students" by Bard Ermentrout (2002).

In addition, the following e-books available through the McMaster library system might be useful:

- "A Primer of Ecology with R" by M. Henry H. Stevens (Springer, 2009)
- "Data Manipulation with R" by Phil Spector (Springer, 2008)
- "Modern Infectious Disease Epidemiology Concepts, Methods, Mathematical Models, and Public Health" edited by Alexander Krämer, Mirjam Kretzschmar and Klaus Krickeberg (Springer, 2010)

Notes

1. **Policy on missed assignments, tests, lectures or tutorials:**
   - `http://www.mcmaster.ca/policy/Students-AcademicStudies/UGCourseMgmt.pdf`.
   - When using the MSAF, the e-mail address to which you should report your absence for Math 4MB3 is `earn@math.mcmaster.ca`. In addition, within two working days, you must also contact the instructor directly by e-mail at `earn@math.mcmaster.ca`. If you miss a test or cannot hand in an assignment on time for a valid reason that has been reported via the MSAF, the final project will then be given appropriate extra weighting. If you must miss a class, it is your responsiblity to find out what was covered. The best way to do this is to borrow a classmate's notes, read them over, and then ask your instructor if there is something that you do not understand.
2. The instructor reserves the right to change the weightings in the grading scheme. If changes are made, your grade will be calculated using the original weightings and the new weightings, and you will be given the higher of the two grades. At the end of the course the grades may be adjusted but this can only increase your grade and will be done uniformly. The McMaster grade equivalence chart will be used to convert between letter grades, grade points and percentages. The grade equivalence chart is published in the Undergraduate Calendar at `http://registrar.mcmaster.ca/CALENDAR/current/pg145.html`
3. No calculators or other aids will be allowed during tests or quizzes unless explicitly indicated.
4. You will be required to bring your official McMaster University photo identification card to the term tests and quizzes.
5. The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in priniciples of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes dishonesty. For information on the various kinds of a academic dishonesty please refer to the Academic Integrity Policy located at `http://www.mcmaster.ca/academicintegrity`. The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g., the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work. In this course, you are encouraged to discuss the assigned problems with other students in your class. However, you must write the solutions in your own words without referring to any other students' work. The copying or even paraphrasing of other students' solutions will be considered academic dishonesty.
3. Copying or using unauthorized aids during tests, quizzes and examinations.

David Earn
23 February 2016

STATS 2MB3: Statistical Methods and Applications
Winter 2016

"Statistics is the science, the art, the philosophy, and the technique of making inferences from the particular to the general," is a well-known quote attributed to John W. Tukey, one of the greatest statisticians of all times. With a strong footing on applications, this course demonstrates the statement repeatedly all along its flow. The course **objectives** are to train the student on: (a) methods to summarize data numerically and graphically, (b) the most widely used statistical methods to draw inferences from observed data, (c) some of the mathematical details behind the methods, and (d) handling data and carrying out basic statistical analyses using the statistical package R.

Course Webpage: We will use **Avenue To Learn** (*http://avenue.mcmaster.ca/*) as the main site for posting course-related material. In Avenue, we will post assignments, midterms, solutions, labs, marks and other relevant information.

Instructor: Dr. Román Viveros-Aguilera, HH/212, ext. 23425, `rviveros@ms.mcmaster.ca`.

Office Hours: Wednesday, Thursday & Friday 9:30-10:30 am.

Lectures: Monday, Wednesday & Thursday 1:30-2:20 pm, HH/109.

Tutorials and Tutor Office Hours: There will be three tutorials to be held as follows: T01: Monday 9:30-10:20 am in HH/102 (Tutor: **Kiret**); T02: Monday 2:30-3:20 pm in HH/102 (Tutor: **Atinder**); and T03: Monday 3:30-4:20 pm in HH/102 (Tutor: **Atinder**). Tutorials will start on **Monday January 11**. Each tutorial will feature solutions on the board to sample problems as well as taking questions from the students. In addition, the tutors will hold office hours at a schedule TBA.

Labs: 2 computer labs will be conducted, the first of them at the end of Chapter 1. The dates for the computer labs will be announced later, at least one week in advance. Bring your laptops for the computer labs and work. In addition, computing work will be done in several in-class presentations when the methods discussed require use of the computer.

Software: The computing work will be done in `R`, a freeware statistical software downloadable for Windows and MacIntosh platforms from **The R Project for Statistical Computing** webpage at `http://www.r-project.org/`. You are strongly advised to download the software at your earliest convenience.

Textbook: *Probability and Statistics for the Engineering and the Sciences*, 9th edition, by Jay L. Devore, Brooks/Cole 2012.

Material to be Covered:

1. Descriptive statistical methods (Ch. 1)
2. Distribution of sample mean and the Central Limit Theorem (Secs. 5.4-5.5)
3. Methods of point estimation (Ch. 6)
4. Interval estimation methods based on single samples (Ch. 7)
5. Tests of hypotheses based on single samples (Ch. 8)
6. Inferences based on two samples (Ch. 9)
7. Simple linear regression (Sections 12.1-12.2)

8. Basic statistical computing in R (class notes and computer labs)

Assignments: There will be six assignments in the course. Make sure you write your complete name and student number on each page of every assignment. You are required to write well-organized and readable solutions. After marking, your assignments will be returned to you, approximately one week after the due date. Solutions to the assignments will be posted in the course webpage. Tentative due dates:

Assignment #1: Wednesday January 20, 3:30 pm.
Assignment #2: Wednesday February 3, 3:30 pm.
Assignment #3: Wednesday February 24, 3:30 pm.
Assignment #4: Wednesday March 9, 3:30 pm.
Assignment #5: Wednesday March 23, 3:30 pm.
Assignment #6: Wednesday April 6, 3:30 pm.

The final due dates will be announced in class. The assignment questions will be posted in the course webpage one week prior to the deadline.

Tests: There will be two midterm tests and a 2.5-hour final exam in the April examinations. The midterms are tentatively scheduled as follows:

Test #1: Wednesday February 10, evening, 1.5 hours, location: TBA.
Test #2: Wednesday March 16, evening, 1.5 hours, location: TBA.

The material covered by the midterm tests will be announced in class. The final exam will be based on all the material covered in the course. Only the McMaster standard calculator Casio fx-991 will be allowed. Solutions to the midterm tests will be posted in the course webpage.

Final Exam: In April, 2.5 hours, scheduled by the Registrar's Office.

Marking Scheme:

	Option 1	**Option 2**
Assignments	10%	10%
Midterm Tests	20%	40%
Final Exam	70%	50%

The final mark will be the highest from **Option 1** and **Option 2**.

Expectations:

(a) **Come to class.** I know you will think that this is an obvious one but it is important nonetheless. Learning statistics is a cumulative experience; what we learn in one class builds upon what was covered in previous sessions. Missing class creates gaps in your knowledge-base that will make hard to build on, and the degree of difficulty will increase rapidly.

(b) **Complete all assignments.** Much like the practice required in learning a foreign language, you will have to practice the techniques we learn in each class to get them to take hold and the homework is a good way to accomplish that. These exercises also will help you familiarize yourself with the software necessary to perform efficiently even simple analyses.

(c) **Ask questions.** Office hours provide a great opportunity to ask for clarifications of the material

and questions about your own attempts to solve a problem. When you are stuck, try to explain someone else (classmate friend, tutor, instructor) what you are trying to do. This can often help you identify for yourself the problem you are having. Often the step we are missing is readily obvious to someone who is not trying to solve the problem.

(d) **Be respectful of others.** Show your respect to your classmates by keeping a quiet and pleasant atmosphere in the classroom so that everyone focus their attention on the presentations and discussions.

These are my secrets for success about learning statistics, they come from my own personal experience as a student and as a teacher of the subject for over 20 years.

Important Message: The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster e-mail and course websites weekly during the term and to note any changes.

Academic Integrity: You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty please refer to the Academic Dishonesty Policy, located at `http://www.mcmaster.ca/academicintegrity`.

The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Absences: Please check

`http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work`

for the most up-to-date information about the **McMaster Student Absence Form** procedures (MSAF). Note that substantive changes were made this year to the procedures.

McMaster University

Department of Mathematics and Statistics

STATISTICS 3A03: Applied Regression Analysis with SAS

Fall 2015

Instructor	Angelo J. Canty
Office	Hamilton Hall 209
Phone	(905) 525-9140 Ext 27079
e-mail	cantya@mcmaster.ca
Course Web Site	http://www.math.mcmaster.ca/canty/teaching/stat3a03
Time of Lectures	Monday 1:30 – 2:20 Wednesday 1:30 – 2:20 Thursday 1:30 – 2:20
Location	T13 125
Office Hours	Monday 12:00–1:00 Thursday 12:00–1:00 or at other times by prior appointment.
Labs	Tuesday 2:30-3:20 (L1) in BSB 244 Friday 2:30–3:20 (L2) in BSB 244
Required Text	*Regression Analysis By Example* (Fifth Edition) by Samprit Chatterjee & Ali S. Hadi. Wiley (2012)
Optional Text (Recommended)	*Applied Linear Regression* (Fourth Edition) by Sanford Weisberg. Wiley (2014)

Grading Scheme :

The final grade will be based on the following components

1. 20% for the best 4 out of 5 assignments.
2. 20% for each of two term tests.
3. 40% for the final exam.

Notes :

- Tentative due dates for the assignments will be at **the start of class** on Sept. 28, Oct. 19, Nov. 2, Nov. 16 and Dec. 7. Assignments will be posted on the web site at least 1 week before the due date.
- The term tests will be held during class time. Tentative dates are October 26 and November 30. The location and other information will be announced in class and on the web site at least one week before the tests.
- The final exam will take place during the December examination period and will cover the entire course. It will be a 2.5 hour exam scheduled by the Examinations Office. Students should check the official exam schedule for the time and location.
- Any student who misses work during the term should complete the online McMaster Student Absence Form (MSAF) within one week of the missed work. Second or subsequent absences must be approved by the associate dean of your faculty. For more details see the website https://www.mcmaster.ca/msaf/index.html. Students using the MSAF should also email the professor a day or two after submitting their request to ensure it was properly received. In the case of a missed term test, the marks for that test will be distributed over the other test (+5%), assignments (+5%) and the final exam (+10%). In the case of one or more missed assignments, the assignment mark will be calculated using **all** other assignments not covered by an MSAF or otherwise approved. **No other modifications of the marking scheme will be considered.**

Academic Dishonesty

Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3, located at
http://www.mcmaster.ca/policy/Students-AcademicStudies/AcademicIntegrity.pdf

The following illustrates only three forms of academic dishonesty:

- **Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.**
- **Improper collaboration in group work.**
- **Copying or using unauthorized aids in tests and examinations**

Brief Outline of the Course

This course will introduce the students to the theory of regression and its implementation using the SAS language. Note that the required textbook is rather applied in nature and does not fully cover the required theory. The optional textbook will be primarily used for the more theoretical parts of the course. The course consists of 3 lectures per week and also one computer lab. The lab is mandatory as this is where the students will learn and be able to use the SAS system. Assignments will require the use of SAS.

The course will commence with a review of some of the important distributions that will come up in this course; primarily the Gaussian (normal), Student's t and chi-squared distributions. We will then examine simple linear regression in which one response is related to a single predictor variable. Most students will have seen this in Stat 2MB3 but in this course we shall delve further into the theoretical basis for simple linear regression. We shall then extend to multiple regression in which there are more than one continuous predictor variables. This part of the course will also involve familiarity with matrix notation and operations.

We will then consider more advanced aspects of regression, the first of these will be diagnostics to assess problems with the underlying assumptions and fitted model. We will follow this by considering the issue of qualitative predictor variables and the concept of interactions. Transformations and weighted least squares are two methods that can overcome some issues in regression and these will be considered next. Finally we shall look at methods to decide on which predictor variables should be included in a final model.

At the end of the course, students should have a reasonable amount of familiarity with the SAS system and be able to use it for data entry and model fitting and checking. They should also understand the theory behind linear models and be able to use them in settings such as regression, analysis of variance and analysis of covariance.

Approximate Timetable

Note that this timetable is subject to change depending on how the class progresses and is intended as a guide only. Major deviations (such as topics being omitted) will be announced in class and on the web site. The textbook sections indicated are the sections most closely related to what we shall cover but actual coverage may not correspond exactly to the textbook.

Dates	**Topic**	**Textbook**
Sept 9 –10	Review and Introduction to Regression	Chapter 1.
Sept 14 – 24	Simple Linear Regression	Chapter 2.
Sept 28 – Oct 8	Multiple Linear Regression	Chapter 3.
Oct 19 – 29	Regression Diagnostics	Chapter 4.
Oct 26	**Term Test 1**	—
Nov 2 – 12	Qualitative Variables in Regression	Chapter 5.
Nov 16 – 19	Transformations	Chapter 6.
Nov 23 – 26	Weighted Least Squares.	Chapter 7.
Nov 30	**Term Test 2**.	—
Dec 2 – Dec 7	Variable Selection	Chapter 11.

Important Notice:

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their **McMaster** email and the course website weekly during the term and to note any changes.

STATS 3D03: Mathematical Statistics – Fall 2014

COURSE OUTLINE

COURSE OBJECTIVES: By the end of the semester, students will have gained the ability to:

- Handle competently the calculation of probabilities, means, variances, covariances and correlations of univariate and multivariate distributions.
- Derive and use the moment generating function of discrete and continuous random variables.
- Establish the convergency in probability and in distribution of basic statistics.
- Derive, calculate and interpret confidence intervals and tests of hypotheses.
- Derive the maximum likelihood estimators for a variety of discrete and continuous distributions.

INSTRUCTOR: Román Viveros-Aguilera, HH/212, ext. 23425, e-mail: `rviveros@math.mcmaster.ca`.

LECTURES: Monday & Wednesday 11:30–12:20 am, Friday 1:30–2:20 pm, room: BSB/120.

OFFICE HOURS: Mondays 2:30–4:00 pm & Wednesdays 9:30–11:00 am.

TEXT: *Introduction to Mathematical Statistics*, 7th Edition, Pearson, 2013.

COURSE WEBPAGE: We will use **Avenue To Learn** (*http://avenue.mcmaster.ca/*) as the main site for posting course-related material. In a needy situation, the alternative site will be `http://icarus.math.mcmaster.ca/rviveros/3d03.html`. In Avenue, we will post some class notes, assignments, midterms, solutions and marks.

COURSE CONTENTS:

1. Probability spaces and distributions (Chapter 1).
2. Multivariate distributions (Chapter 2).
3. Important families of distributions (Chapter 3).
4. Statistical inferences: confidence intervals and tests of hypotheses (Chapter 4).
5. Limiting distributions (Chapter 5).
6. Estimation: maximum likelihood (Chapter 6, time permitting).

ASSIGNMENTS: There will be six assignments tentatively scheduled as follows:

Assignment #1 – Due-date: **4:00 PM on Wednesday September 17.**
Assignment #2 – Due-date: **4:00 PM on Wednesday October 1.**
Assignment #3 – Due-date: **4:00 PM on Wednesday October 15.**
Assignment #4 – Due-date: **4:00 PM on Wednesday October 29.**

Assignment #5 – Due-date: **4:00 PM on Wednesday November 12.**

Assignment #6 – Due-date: **4:00 PM on Wednesday November 26.**

Final due-dates will be confirmed in class. Each assignment will be posted in the course webpage one week prior to its due-date. Your assignment papers should be dropped-off in the course's designated locker, located near room 105 in Hamilton Hall, please do not drop-off papers under the instructor's office door. Your assignment papers are expected to be clearly and sequentially written, well organized and with each part properly labeled. To be fair to all, no late assignments will be accepted.

MIDTERM TESTS: There will be two midterm tests tentatively scheduled as follows:

Midterm #1: **Evening of Wednesday October 8,** room: TBA.

Midterm #2: **Evening of Wednesday November 5,** room: TBA.

Final confirmation will be announced in class.

FINAL EXAM: A 3-hour final exam will be held in the December regular exam period. Scheduling will be done by the Registrar.

CALCULATORS: The standard Casio fx-991 calculator will be the only calculator allowed for midterm and final exam.

MARKING SCHEME:

	Option 1	**Option 2**	**Option 3**
Assignments	10%	10%	0%
Midterm Tests	20%	40%	0%
Final Exam	70%	50%	100%

The final mark will be the highest from **Option 1**, **Option 2** and **Option 3**.

EXPECTATIONS:

(a) **Come to class.** I know you will think that this is an obvious one but it is important nonetheless. Learning statistics is a cumulative experience; what we learn in one class builds upon what was covered in previous sessions. Missing class creates gaps in your knowledge base that will make hard to build on, and the degree of difficulty will increase rapidly.

(b) **Complete all assignments.** Much like the practice required in learning a foreign language, you will have to practice the techniques we learn in each class to get them to take hold and the homework is a good way to accomplish that. These exercises also will help you familiarize yourself with the software necessary to perform efficiently even simple analyses.

(c) **Ask questions.** Office hours provide a great opportunity to ask for clarifications of the material and questions about your own attempts to solve a problem. When you are stuck, try to explain someone else (classmate friend, tutor, instructor) what you are trying to do. This can often help you identify for yourself the problem you are having. Often the step we are missing is readily obvious to someone who is not trying to solve the problem.

(d) **Be respectful of others.** Show your respect to your classmates by keeping a quiet and pleasant atmosphere in the classroom so that everyone focus their attention on the presentations and discussions.

These are my secrets for success in learning statistics, they come from my own personal experience as a student and as a teacher of the subject for over 20 years.

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ACADEMIC INTEGRITY: You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

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2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

ABSENCES: If you are absent from the university for a minor medical reason, lasting fewer than 5 days, you may report your absence, once per term, without documentation, using the McMaster Student Absence Form (MSAF, *http://mcmaster.ca/msaf/*). For the most up-to-date MSAF information, found in the 2014-2015 Undergraduate Calendar, visit *http://academiccalendars.romcmaster.ca/content.php?catoid=7&navoid=559#Requests_for_Relief_for_Missed_Academic_Term_Work.* Absences for a longer duration or for other reasons must be reported to your Faculty/Program office, with documentation, and relief from term work may not necessarily be granted. When using the MSAF, report your absence to the instructor immediately (normally within 2 working days) by e-mail to `rviveros@math.mcmaster.ca`. The course policy for an approved request for missed work is: for a missed assignment, the weight will be equally distributed on the other assignments; for a missed midterm, the weight will be added proportionately to the remaining test and final exam. Please note that the MSAF may not be used for term work worth 30% or more, nor can it be used for the final examination.

STATS 3G03 Actuarial Mathematics I

Information Sheet (Fall 2015)

LECTURE	**Thursday 7:00-10:00 pm**	**HH/109**
	Classes:	September 8 -December 8
	Mid-term recess:	October 12-17
	Test/Exam ban:	December 2-8

CLASS MATERIAL: Available through Avenue to Learn **(http://avenue.mcmaster.ca/)**

Instructor	Margaret Verbeek
Office	HH/425
Phone	905-525-9140 x 23408
Email	verbeem@mcmaster.ca
Office Hours	Wednesday 1230-130pm, Thursday 6:00-7:00pm or by appt

Assignment Drop Boxes: TBA (based on last name), next to HH/105

Tutorial Name Feng Tian Email: tianf2@math.mcmaster.ca Office: HH/301
Tuesday 10:30-11:20am HH/104
Starting week of September 14th

Textbook:

Actuarial Mathematics for Life Contingent Risks, 2nd Edition, by David CM Dickson, Mary R Hardy, and Howard R Waters

Optional: Solutions Manual for Actuarial Mathematics for Life Contingent Risks (2nd Edition) by David CM Dickson, Mary R Hardy, and Howard R Waters

Optional Reference Material:

Models for Quantifying Risk, 4th or 5th edition, by Cunningham R, Herzog T, and London R, published by Actex and Solutions Manual for Models for Quantifying Risk (matching edition)

Edu-2013-spring-mlc-supp-mrq5a.pdf (available on SOA website)

Course Description:

Introduction to Actuarial Mathematics covering survival distributions, life tables, life insurance, life annuities, and premiums. The material, together with course 3H03, will form the basis of the material needed for Society of Actuaries Actuarial Exam MLC.

Prerequisite for the course is Statistics 2D03 and one of Math 2FM3 or 2K03.

Lectures:

There will be 1 lecture per week, 3 hours in length. A list of supporting material (including sample problems) will be posted on the course webpage. It is important to attempt as many of the supporting problems as you can. This is essential to the learning process. Answers to some supporting problems can be found in the textbook. There is a one hour tutorial for this course.

Assignments:

There will be approximately 3 assignments during the term. Assignments are to be submitted in the lockers on the first floor (basement) of Hamilton Hall *(near HH/105 – box C52 & C53)*.

Tentative schedule is as follows:

Assignment 1	Sunday Sep 20	- due Monday Sep 28
Assignment 2	Tuesday Oct 13	- due Friday Oct 23
Assignment 3	Friday Nov 13	- due Friday Nov 20

Tests:

There will be three 1.5 hour tests tentatively set for:

October 1	Time 7-830pm	Location HH102 & HH109
October 29	Time 7-830pm	Location HH102 & HH109
November 26	Time 7-830pm	Location HH102 & HH109

The topics covered on the test will be announced on the course webpage. The date and time of the tests are not negotiable. If you have conflicts with any of these times, inform M. Verbeek by email within the first two weeks of class. Include the nature of the conflict, and your availability earlier on the test day as well as the next day.

You must bring your student ID to each test.

The standard McMaster calculator Casio fx991 must be used during tests.

Note that the midterm recess will be Monday October 12 to Saturday October 17. There will be no class or test on these days.

Final Examination:

The final examination (duration 2.5 hours) will be scheduled by the registrar.

Marking Scheme:

Final examination	50%
Mid Term tests	30%
Assignments	20%

Official Policies:

1. **Exemptions from course work –** Students seeking an exemption for an assignment, project or mid-term test must contact the office of the Associate Dean (Studies) of their Faculty within a week from the due date of the assignment or project or the test date with proper documentation, e.g. a doctor's note. The section instructor will act on the exemption only after receiving official communication from the relevant Associate Dean's office that the grounds for exemption have been accepted. Deferrals for the final exam are handled exclusively by the Associate Dean's office. The percentages for exempted assignments, projects and tests will be transferred to the final exam.
2. **Late assignments and projects will not be graded.**

3. **Calculators** – Only the McMaster standard calculator, the Casio fx991, may be used during the tests and the final exam.
4. **ID cards** – Students must bring their McMaster ID card to every test ad to the final exam for inspection.
5. **Academic ethics** – You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

   Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty") and/or suspension or expulsion from the university.

   It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of academic dishonesty please refer to the Academic Integrity Policy located at http://www.mcmaster.ca/academicintegrity.

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   b. Improper collaboration in group work.
   c. Copying or using unauthorized aids in tests and examinations.

6. **Academic Accommodation for Students –** Students who require academic accommodation must contact Student Accessibility Services (SAS) to make arrangements with a Program Coordinator. Academic accommodations must be arranged for each term of study. Student Accessibility Services can be contacted by phone 905-525-9140 x 28652 or e-mail sas@mcmaster.ca. For further information, consult McMaster University's Policy for Academic Accommodation of Students with Disabilities.

7. **Requests for Relief for Missed Academic Term Work -** In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work". Please note these regulations have changed beginning Fall 2015. A copy of the updated regulations is available on A2L. See also **http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work**

Important Message

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STATS3H03 Actuarial Mathematics II

Information Sheet (Winter 2016)

LECTURE	**Wednesday 7:00-10:00 pm**	**HH/109**
	Classes:	January 5 to April 8, inclusive
	Mid-term recess:	February 15-20
	Test/Exam ban:	April 4-11

CLASS MATERIAL: Available through Avenue to Learn **(http://avenue.mcmaster.ca/)**

Instructor	Margaret Verbeek
Office	HH/425
Phone	905-525-9140 x 23408
Email	verbeem@mcmaster.ca
Office Hours	Wednesday 6:00-7:00pm or by appt

Assignment Drop Boxes: #63, 64 (based on last name), next to HH/105

Tutorial Name Feng Tian Email: tianf2@math.mcmaster.ca Office: HH/301
Monday 9:30-10:20am HH/104
Starting week of January 11th

Textbook:

Actuarial Mathematics for Life Contingent Risks, 2nd Edition, by David CM Dickson, Mary R Hardy, and Howard R Waters

Optional: Solutions Manual for Actuarial Mathematics for Life Contingent Risks (2nd Edition) by David CM Dickson, Mary R Hardy, and Howard R Waters

Optional Reference Material:

Models for Quantifying Risk, 4th or 5th edition, by Cunningham R, Herzog T, and London R, published by Actex and Solutions Manual for Models for Quantifying Risk (matching edition)

Edu-2013-spring-mlc-supp-mrq5a.pdf (available on SOA website)

Course Description:

Introduction to Actuarial Mathematics II will cover policy values, multiple state models, multiple decrements, multiple lives, pensions, participating and universal life products, and yield curves. The material, together with course 3G03, will form the basis of the material needed for Society of Actuaries Actuarial Exam MLC.

Prerequisite for the course is Statistics 3G03.

Lectures:

There will be 1 lecture per week, 3 hours in length. A list of supporting material (including sample problems) will be posted on the course webpage. It is important to attempt as many of the supporting problems as you can. This is essential to the learning process. Answers to some supporting problems can be found in the textbook. There is a one hour tutorial for this course.

Assignments:

There will be approximately 3 assignments during the term. Assignments are to be submitted in the lockers on the first floor (basement) of Hamilton Hall *(near HH/105 – box C52 & C53)*.

Tentative schedule is as follows:

Assignment 1	January 15th	- due January 22
Assignment 2	February 12th	- due February 26
Assignment 3	March 11th	- due March 18

Tests:

There will be three 1.5 hour tests tentatively set for:

January 27	Time 7-830pm	Location T13 rooms 106, 127
March 2	Time 7-830pm	Location HH102/HH109
March 30	Time 7-830pm	Location HH102/HH109

The topics covered on the test will be announced on the course webpage. The date and time of the tests are not negotiable. If you have conflicts with any of these times, inform M. Verbeek by email within the first two weeks of class. Include the nature of the conflict, and your availability earlier on the test day as well as the next day.

You must bring your student ID to each test.

The standard McMaster calculator Casio fx991 must be used during tests.

Note that the midterm recess will be Monday February 15th to Saturday February 20. There will be no class or test on these days.

Final Examination:

The final examination (duration 2.5 hours) will be scheduled by the registrar.

Marking Scheme:

Final examination	50%
Mid Term tests	30%
Assignments	20%

Official Policies:

1. **Exemptions from course work –** Students seeking an exemption for an assignment, project or mid-term test must contact the office of the Associate Dean (Studies) of their Faculty within a week from the due date of the assignment or project or the test date with proper documentation, e.g. a doctor's note. The section instructor will act on the exemption only after receiving official communication from the relevant Associate Dean's office that the grounds for exemption have been accepted. Deferrals for the final exam are handled exclusively by the Associate Dean's office. The percentages for exempted assignments, projects and tests will be transferred to the final exam.
2. **Late assignments and projects will not be graded.**
3. **Calculators** – Only the McMaster standard calculator, the Casio fx991, may be used during the tests and the final exam.
4. **ID cards** – Students must bring their McMaster ID card to every test ad to the final exam for inspection.

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Important Message

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STATISTICS 3PG03 COURSE OUTLINE
2015-2016 (TERM 2)

INSTRUCTOR: Dr. Fred M. Hoppe, HH/205, (905) 525-9140 Ext. 24688. Email: hoppe@mcmaster.ca. Office hours: Tues. 10:30 – 11:20, Fri. 11:30 – 12:20.

TA: Kai Liu (liuk25@math.mcmaster.ca)

LECTURES: Tues., Wed., Fri. 9:30 am in HH/109.

TEXT: The Doctrine of Chances: Probabilistic Aspects of Gambling, by Stewart N. Ethier, Publisher: Springer, ISBN 978-354078782. There is much more than even a year's worth of material in this book and only some selected chapters will be covered. A strong background in probability is a prerequisite.

CALENDAR DESCRIPTION: Conditional expectation; discrete martingales, Markov chains; game theory; house advantage, gambler's ruin, betting systems, bold play, optimal proportional play and card theory; probabilistic treatment of games of chance. *Note that this is not a course to teach you how to gamble but rather a serious course in applied probability in which the motivation and application is based on gambling, the topic that motivated some of the earliest published works in the area.*

COURSE WEBSITE: Some material will be available on Avenue. Students are expected to check Avenue regularly for postings, assignments, and news about the course.

ASSESSMENT: The final grade will be based on assignments (70%) and a cumulative final examination (30%). The letter grade will be determined from the final percent using the grade equivalence chart in the Undergraduate Calendar. The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of all students to check their McMaster email and Avenue prior to each class to note any changes. The instructor also reserves the right to change the weight of any portion of the grading scheme in such a way that if changes are made, each student's grade will be the higher of the original weightings and the new weightings. At the end of the course the grades may be adjusted but this can only increase the grades and will be done uniformly.

CALCULATOR: McMaster University endorses a "standard" calculator (Casio fx991). This is the only calculator permitted on tests and the final examination.

POLICY FOR MISSED WORK: A missed/late assignment or a missed test counts as 0%. However, you should be aware of the policy (see below) regarding the McMaster

Student Absence Form (MSAF). The weight of any excused work from MSAF will be added to the final exam.

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work." Please note that these regulations have changed beginning Spring/Summer 2015. If you have any questions about the MSAF, contact your Associate Dean's office.

McMaster Policy on Academic Integrity: You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, such as a grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty, please refer to the Academic Integrity Policy, at http://www.mcmaster.ca/academicintegrity. The following illustrates only three forms of academic dishonesty: 1) Plagiarism—e.g., the submission of work that is not one's own or for which other credit has been obtained; 2) Improper collaboration in group work; 3) Copying or using unauthorized aids in tests and examinations.

Academic Accommodation of Students with Disabilities: Students who require academic accommodation must contact Student Accessibility Services (SAS) to make arrangements with a Program Coordinator. Academic accommodations must be arranged for each term of study. SAS can be contacted by phone (905) 525-9140 Ext. 28652 or email (sas@mcmaster.ca). For further information, consult McMaster University's Policy for Academic Accommodation of Students with Disabilities.

Email Contact and Student Responsibility Statement: Please note: The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. The instructor also reserves the right to change the weight of any portion of the grading scheme in such a way that if changes are made, each student's grade will be the higher of the original weightings and the new weightings. It is the responsibility of students to check the course website and their McMaster email weekly during the term and to note any changes. Announcements will be made in class and on the course website.

STATS 3S03: Survey Sampling
Fall 2015

This course introduces students to some of the principles and procedures involved in the planning and conduction of a sample survey, and in the corresponding data analysis. Emphasis is placed on the merits, appropriateness and properties of a variety of well-known probability sampling schemes for a finite population, and on the associated statistical analyses. The main sampling designs studied in the course are simple random sampling, systematic sampling, stratified sampling, cluster sampling and unequal probability sampling. The topics will be discussed with reference to sample surveys from a variety of applications.

Course Webpage: We will use **Avenue To Learn** (*http://avenue.mcmaster.ca/*) as the main site for posting course-related material. In Avenue, we will post assignments, midterms, solutions, marks and other relevant information.

Instructor: Dr. Román Viveros-Aguilera, HH/212, ext. 23425, `rviveros@ms.mcmaster.ca`.

Office Hours: Monday 2:30–4:00 pm, and Tuesday & Wednesday 10:00–11:00 am.

Lectures: Tuesday, Wednesday & Friday 12:30-1:20 pm, HH/104.

Software: The computing work will be done in `R`, a freeware statistical software downloadable for Windows and MacIntosh platforms from **The R Project for Statistical Computing** webpage at `http://www.r-project.org/`. The computing will be integrated with the course.

Textbook: *Sampling: Design and Analysis*, 2nd edition, by Sharon Lohr, Brooks/Cole 2009.

Material to be Covered:

1. Basic ideas about sampling (Ch. 1)
2. Simple random sampling and systematic sampling, and estimation (Ch. 2)
3. Stratified sampling and estimation, and Neyman allocation formula (Ch. 3)
4. Ratio and regression estimation (Ch. 4)
5. Cluster sampling and estimation (Ch. 5)
6. Unequal probability sampling and estimation (Ch 6)
7. Complex surveys and estimation (Ch. 7, time permitting)

Assignments: There will be six assignments in the course. Make sure you write your complete name and student number on each page of every assignment. You are required to write well-organized and readable solutions. After marking, your assignments will be returned to you, approximately one week after the due date. Solutions to the assignments will be posted in the course webpage. Tentative due dates:

Assignment #1: Tuesday September 22, 3:30 pm
Assignment #2: Tuesday October 6, 3:30 pm
Assignment #3: Tuesday October 27, 3:30 pm

Assignment #4: Tuesday November 10, 3:30 pm
Assignment #5: Tuesday November 24, 3:30 pm
Assignment #6: Tuesday December 1, 3:30 pm

The final due dates will be announced in class. The assignment questions will be posted in the course webpage one week prior to the deadline.

Tests: There will be two midterm tests and a 2.5-hour final exam in the December examinations. The midterms are tentatively scheduled as follows:

Test #1: Tuesday October 20, class time, 50 minutes, location: TBA.
Test #2: Tuesday November 17, class time, 50 minutes, location: TBA.

The material covered by the midterm tests will be announced in class. The final exam will be based on all the material covered in the course. Only the McMaster standard calculator Casio fx-991 will be allowed. Solutions to the midterm tests will be posted in the course webpage.

Final Exam: In December, 2.5 hours, scheduled by the Registrar's Office.

Marking Scheme:

	Option 1	**Option 2**	**Option 3**
Assignments	10%	10%	0%
Midterm Tests	20%	40%	0%
Final Exam	70%	50%	100%

The final mark will be the highest from **Option 1**, **Option 2** and **Option 3**.

Expectations:

(a) **Come to class.** Learning statistics is a cumulative experience; what we learn in one class builds upon what was covered in previous sessions. Missing class creates gaps in your knowledge base that will make hard to build on, and the degree of difficulty will increase rapidly.

(b) **Complete all assignments.** Much like the practice required in learning a foreign language, you will have to practice the techniques we learn in each class to get them to take hold and the homework is a good way to accomplish that. These exercises also will help you familiarize yourself with the software necessary to perform efficiently even simple analyses.

(c) **Ask questions.** Office hours provide a great opportunity to ask for clarifications of the material and questions about your own attempts to solve a problem. When you are stuck, try to explain someone else (classmate friend, tutor, instructor) what you are trying to do. This can often help you identify for yourself the problem you are having. Often the step we are missing is readily obvious to someone who is not trying to solve the problem.

(d) **Be respectful of others.** Show your respect to your classmates by keeping a quiet and pleasant atmosphere in the classroom so that everyone focus their attention on the presentations and discussions.

These are my secrets for success in learning statistics, they come from my own personal experience as a

student and as a teacher of the subject for over 20 years.

Important Message: The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Academic Integrity: You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty please refer to the Academic Dishonesty Policy, located at `http://www.mcmaster.ca/academicintegrity`.

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1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Abcenses: Please check

`http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work`

for the most up-to-date information about the **McMaster Student Absence Form** procedures (MSAF). Note that substantive changes were made this year to the procedures.

STATS 3U03

Term 2, 2013/14

Instructor

- **Dr. M. Min-Oo, HH 313, ext. 23429**; e-mail: minoo@mcmaster.ca
- home page: http:// www.math.mcmaster.ca/minoo

- Office hours: **Monday, Wednesday and Thursday: 17:35 - 18:50**
-

Lectures

- **Mondays, Wednesdays and Thursdays 16:30 - 17:20 in Hamilton Hall 102**

Course Description

Description from Undergraduate Calendar
Random walk, Markov chains, discrete and continuous parameter Markov processes, branching processes, birth and death processes, queuing processes.

My description
Discrete Time Markov Chains, Continuous Time Markov Chains, Renewal Processes, Optimal Stopping
plus selected topics from: Martingales, Reversible Markov Chains, Brownian Motion and Stochastic Integration (if time permits)

Required Textbook:
"Introduction to Stochastic Processes", 2nd edition, by Gregory F. Lawler, published by Chapman & Hall CRC Press (the first edition is also fine by me)

Other introductory textbooks:
"Introduction to Stochastic Processes" by Hoel Port Stone
"Probability Models" by Sheldon Ross
"An Introduction to Stochastic Modelling" by Howard Taylor and Samuel Karlin
"Markov Chains" by James Norris

slightly more advanced:
"A first course in Stochastic Processes" by Samuel Karlin and Howard Taylor
"Stochastic Processes and Models" by David Stirzaker

Course Objective:
We will cover the material from Chapters 1 to 6 and selected sections from Chapters 7 to 9. I also plan to add some supplementary material which is not in the required textbook.

Students are responsible for reading the relevant material from the textbook and/or the reference books and/or any other resources (freely available on the internet) and also for working out most of the exercises in the book ***on their own*** *(that's known as experiential learning) in preparation for the lectures*

For a weekly update on what is covered in the course see the **course syllabus**

Course Work:

1. Assignments: There will be five written assignments to be handed in on the due dates that will be announced in class. Late assignments will not be graded.

Doing the assignments regularly is the key to understanding the material

- Assignment #1 due: in class on Wednesday, January 22nd click here for short answers to selected questions
- Assignment #2 due: in class on Wednesday, February 5th click here for short answers to selected questions
- Assignment #3 due: in class on Thursday, March 6th click here for short answers to selected questions
- Assignment #4 due: in class on Thursday, March 20th click here for short answers to selected questions
- Assignment #5 due: in class on Wednesday, April 2nd

2. Homework: In addition to the assignments, there will be extra homework problems from the textbook which are not graded. Click here for Homework..

3. Tests: There will be two 45 minute tests done in class. The exact dates will be announced in class and this course home page.

- **TEST #1 Thursday, February 13th in class during the lecture period**
- **TEST #2 Thursday, March 27th in class during the lecture period**

4. Final Examination: There will be a 3-hour final examination scheduled by the Registrar's office during the exam period in December.

Grading Scheme

- Assignments: 30%
- Tests: 25%
- Final Exam: 45%

Academic Dishonesty:

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The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.

2. Improper collaboration in group work.

3. Copying or using unauthorized aids in tests and examinations.

Other Policy Notes:

MSAF policy: (see: http://registrar.mcmaster.ca/CALENDAR/2013-14/pg2246.html)

When using the MSAF, also report your absence to me (the course instructor M. Min-Oo) immediately (normally within 2 working days) by email (minoo@mcmaster.ca) and contact me in person to learn what relief may be granted for the work you have missed, and relevant details such as revised deadlines, or time and location of a make-up exam. Please note that the MSAF may not be used for term work worth 30% or more, nor can it be used for the final examination. Here are the rules:

1. Requests for Relief for Missed Academic Term Work for absences from classes lasting up to 5 days due to a minor medical situation: Using the McMaster Student Absence Form (MSAF) on-line self-reporting tool, undergraduate students may report absences due to minor medical situations lasting up to 5 days and may also request relief for missed academic work worth less than 30% of the final grade. The submission of medical documentation is normally not required. Students may use this tool to submit a maximum of one request for relief of missed academic work per term. Students must immediately (e.g. by e-mail within 2 days) follow up with their course instructors regarding the nature of the relief. Failure to do so may negate the opportunity for relief. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work. relief for missed term work.

2. For absences from classes lasting more than 5 days: Students who are absent more than five days cannot use the on-line, self-reporting tool to request relief. They MUST report to their Faculty Office to discuss their situation and may be required to provide appropriate supporting documentation. If warranted, students will be approved to use a discretionary version of the MSAF on-line, self-reporting tool.

3. For the reporting of more than one request for relief per term: Students who wish to submit more than one request for relief of missed academic work per term cannot use the on-line, self-reporting tool to request relief. They MUST report to the office office of the Associate Dean (Studies) of their Faculty within a week of the test date, and may be required to provide appropriate supporting documentation, e.g. a doctor's note. If warranted, students will be approved to use a discretionary version of the MSAF on-line, self-reporting tool. The instructor will act on the exemption only after receiving official communication from the relevant Associate Dean's office that the grounds for exemption have been accepted.

4. For absences from classes unrelated to a medical situation: Students who are absent for reasons other than a medical situation cannot use the on-line, self-reporting tool to request relief. They MUST report to their Faculty Office to discuss their situation and may be required to provide appropriate supporting documentation. If warranted, students will be approved to use a discretionary version of the MSAF on-line, self reporting tool.

5. For relief from missed work worth 30% or more of the final grade: Students who have missed a piece of work worth 30% or more cannot use the on-line, self-reporting tool to request relief. They MUST report to their Faculty Office to discuss their situation and may be required to provide appropriate supporting documentation. If warranted, students will be approved to use a discretionary version of the MSAF on-line, self-reporting tool.

6. For absences from classes lasting more than five days, for work worth 30% or more, or for the reporting of more than one request for relief per term: If the reason was medical, the approved McMaster University Medical Form covering the relevant dates must be submitted. The student must be seen by a doctor at the earliest possible date, normally on or before the date of the missed work and the doctor must verify the duration of the illness. Relief will not be considered for minor illnesses. If the reason is non-medical, appropriate documentation with verifiable origin covering the relevant dates must be submitted, normally within five working days. In some circumstances, students may be advised to submit a Petition for Special Consideration (Form A) seeking relief for missed academic work. In deciding whether or not to grant a petition, adequacy of the supporting documentation, including the timing in relation to the due date of the missed work and the degree of the student's incapacitation, may be taken into account. If the petition is approved the Faculty Office will notify the instructor(s) recommending relief. The student must contact the instructor promptly to discuss the appropriate relief. Failure to do so may negate the opportunity for relief. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work.

7. The MSAF on-line, self-reporting tool cannot be used to apply for relief for any final examination or its equivalent. Deferrals for the final exam are handled exclusively by the Associate Dean's office.

Calculators:

Only the standard McMaster calculator Casio fx 991MS+ can be used for the tests and the final examination.

Important Notice:

The instructor and the university reserve the right to modify or revise information contained in this course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification or revision becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Tentative schedule of Topics

(the numbers are chapters and sections from the text book)

Week 1 (06/01 to 10/01): 1.1, 1.2, 1.3, 1.4

Week 2 (13/01 to 17/01): 1.5, 1.6, 2.1, 2.2

Week 3 (20/01 to 24/01): 2.2, 2.3, 2.4

Week 4 (27/01 to 31/01): 3.1, 3.2

Week 5 (03/02 to 07/02): 3.3, 3.4

Week 6 (10/02 to 14/02): Chapter 4 (Optimal Stopping)

Week 7 (17/02 to 21/02) READING WEEK ;)

Week 8 (24/02 to 28/02): Chaper 6 (Renewal processes)

Week 9 (03/03 to 07/03): selected material from Chapter 5 (Martingales)

Week 10 (10/03 to 14/03): selected material from Chapter 7 (Reversible Markov Chains)

Week 11 (17/03 to 21/03): selected material from Chapter 8 (Brownian Motion)

Week 12 (24/03 to 28/03): selected material from Chapter 9 (Stochastic Integration)

Week 13 (31/03 to 04/04): some extra material (such as MCMC)

Week 14 (07/04 to 08/04) Review

STATS 4A03/6A03: Time Series – Winter 2016
COURSE OUTLINE

Course Objectives: Upon successful completion of this course, you will be able to perform the following tasks:

- Produce, interpret and explain time plots, including identification of trends and seasonality, for a time series dataset.
- Calculate, plot and interpret autocorrelation functions.
- Fit and interpret time series models such as ARMA (autoregressive moving average) and ARIMA (autoregressive integrated moving average) to time series data.
- Carry out forecasts from time series data and interpret and explain the results.
- State and justify mathematically the main theoretical tools, apply the methods numerically to datasets, and interpret and explain the results.
- Handle the numerical calculations and plotting in the computer using statistical software such as R.

Course Webpage: We will use **Avenue To Learn** (*http://avenue.mcmaster.ca/*) as the main site for posting course-related material. In Avenue, we will post assignments, midterms, solutions, labs, marks and other relevant information.

Instructor: Dr. Román Viveros-Aguilera, HH/212, ext. 23425, `rviveros@ms.mcmaster.ca`.

Office Hours: Wednesday, Thursday & Friday, 9:30-10:30 am.

Lectures: Monday & Wednesday 8:30-9:20 am, Friday 10:30-11:20 am, HH/305.

Software: The computing work will be done in `R`, a freeware statistical software downloadable for Windows and MacIntosh platforms from **The R Project for Statistical Computing** webpage at `http://www.r-project.org/`. You are strongly advised to download the software at your earliest convenience.

Textbook: *Time Series Analysis With Applications in R*, 2nd edition, by Jonathan D. Cryer & Kung-Sik Chan, Springer 2008.

Material to be Covered:

1. Introductory Examples (Ch. 1)
2. The Main Concepts (Ch. 2)
3. Trends in Time Series (Ch. 3)
4. Models for Stationary Time Series (Ch. 4)
5. Models for Nonstationary Time Series (Ch. 5)
6. Model Specification (Ch. 6)
7. Parameter Estimation for Time Series (Ch. 7)
8. Model Diagnostics (Ch. 8)
9. Forecasting (Ch. 9)

Assignments: There will be six assignments in the course. Make sure you write your complete name and student number on each page of every assignment. You are required to write well-organized and readable solutions. After marking, your assignments will be returned to you, approximately one week after the due date. Solutions to the assignments will be posted in the course webpage. Tentative due dates:

Assignment #1: Monday January 18, 3:30 pm.
Assignment #2: Monday February 1, 3:30 pm.
Assignment #3: Monday February 22, 3:30 pm.
Assignment #4: Monday March 7, 3:30 pm.
Assignment #5: Monday March 21, 3:30 pm.
Assignment #6: Monday April 4, 3:30 pm.

The final due dates will be announced in class. The assignment questions will be posted in the course webpage one week prior to the deadline.

Tests: There will be two midterm tests and a 2.5-hour final exam in the April examinations. The midterms are tentatively scheduled as follows:

Test #1: Monday February 8, evening, 1.5 hours, location: TBA.
Test #2: Monday March 14, evening, 1.5 hours, location: TBA.

The material covered by the midterm tests will be announced in class. The final exam will be based on all the material covered in the course. Only the McMaster standard calculator Casio fx-991 will be allowed. Solutions to the midterm tests will be posted in the course webpage.

Final Exam: In April, 2.5 hours, scheduled by the Registrar's Office.

Marking Scheme:

	Option 1	**Option 2**
Assignments	10%	10%
Midterm Tests	20%	40%
Final Exam	70%	50%

The final mark will be the highest from **Option 1** & **Option 2**.

Expectations:

(a) **Come to class.** I know you will think that this is an obvious one but it is important nonetheless. Learning statistics is a cumulative experience; what we learn in one class builds upon what was covered in previous sessions. Missing class creates gaps in your knowledge base that will make hard to build on, and the degree of difficulty will increase rapidly.

(b) **Complete all assignments.** Much like the practice required in learning a foreign language, you will have to practice the techniques we learn in each class to get them to take hold and the homework is a good way to accomplish that. These exercises also will help you familiarize yourself with the software necessary to perform efficiently even simple analyses.

(c) **Ask questions.** Office hours provide a great opportunity to ask for clarifications of the material and questions about your own attempts to solve a problem. When you are stuck, try to explain someone else (classmate friend, tutor, instructor) what you are trying to do. This can often help you identify for yourself the problem you are having. Often the step we are missing is readily obvious to someone who is not trying to solve the problem.

(d) **Be respectful of others.** Show your respect to your classmates by keeping a quiet and pleasant atmosphere in the classroom so that everyone focus their attention on the presentations and discussions.

These are my secrets for success about learning statistics, they come from my own personal experience as a student and as a teacher of the subject for over 20 years.

Important Message: The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster e-mail and course websites weekly during the term and to note any changes.

Academic Integrity: You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty please refer to the Academic Dishonesty Policy, located at `http://www.mcmaster.ca/academicintegrity`.

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1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Absences: Please check

`http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work`

for the most up-to-date information about the **McMaster Student Absence Form** procedures (MSAF). Note that substantive changes were made this year to the procedures.

STATS 4C03/6C03: Generalized Linear Models
Fall 2015

Generalized linear models is a collection of models and methods useful to describe the statistical relationship between a response variable (outcome or dependent variable) and one or several explanatory variables (predictors or covariates or independent variables). Response and predictor variables can be numerical, ordinal or categorical. The actual relationship is modeled through a so-called link function. Important particular cases are normal multiple linear regression (studied in detail in STATS 3A03: Applied Regression Analysis with SAS), Poisson regression and logistic regression. Special attention will be given to these particular cases, with a focus on understanding the models (i.e. model interpretation), estimation of model parameters (mostly using maximum likelihood), carrying out model fitting in `R`, and applications.

Course Webpage: We will use **Avenue To Learn** (*http://avenue.mcmaster.ca/*) as the main site for posting course-related material. In Avenue, we will post assignments, midterms, solutions, marks and other relevant information.

Instructor: Dr. Román Viveros-Aguilera, HH/212, ext. 23425, `rviveros@ms.mcmaster.ca`.

Office Hours: Monday 2:30–4:00 pm, and Tuesday & Wednesday 10:00–11:00 am.

Lectures: Monday & Wednesday 8:30–9:20 am, & Friday 10:30-11:20 am, HH/102.

Software: The computing work will be done in `R`, a freeware statistical software downloadable for Windows and MacIntosh platforms from **The R Project for Statistical Computing** webpage at `http://www.r-project.org/`. The computing will be integrated with the course.

Textbook: *An Introduction to Generalized Linear Models*, 3rd edition, by Annette J. Dobson and Adrian G. Barnett, CRC Press 2008.

Material to be Covered:

1. Introduction: Review of some distribution theory (Ch. 1)
2. Model fitting (Ch. 2)
3. Exponential family and generalized linear models (Ch. 3)
4. Parameter estimation (Ch. 4)
5. Inferential methods (Ch. 5)
6. Normal linear models(mostly a review from STATS 3A03) (Ch. 6)
7. Binary responses and logistic regression (Ch 7)
8. Poisson regression and log-linear models (Ch. 9)
9. Survival analysis (Ch. 10, time permitting)

Assignments: There will be six assignments in the course. Make sure you write your complete name and student number on each page of every assignment. You are required to write well-organized and readable solutions. After marking, your assignments will be returned to you, approximately one week after the due date. Solutions to the assignments will be posted in the course webpage. Tentative due

dates:

Assignment #1: Wednesday September 23, 3:30 pm
Assignment #2: Wednesday October 7, 3:30 pm
Assignment #3: Wednesday October 28, 3:30 pm
Assignment #4: Wednesday November 11, 3:30 pm
Assignment #5: Wednesday November 25, 3:30 pm
Assignment #6: Wednesday December 2, 3:30 pm

The final due dates will be announced in class. The assignment questions will be posted in the course webpage one week prior to the deadline.

Tests: There will be two midterm tests and a 2.5-hour final exam in the December examinations. The midterms are tentatively scheduled as follows:

Test #1: Wednesday October 21, class time, 50 minutes, location: TBA.
Test #2: Wednesday November 18, class time, 50 minutes, location: TBA.

The material covered by the midterm tests will be announced in class. The final exam will be based on all the material covered in the course. Only the McMaster standard calculator Casio fx-991 will be allowed. Solutions to the midterm tests will be posted in the course webpage.

Final Exam: In December, 2.5 hours, scheduled by the Registrar's Office.

Marking Scheme:

	Option 1	**Option 2**	**Option 3**
Assignments	10%	10%	0%
Midterm Tests	20%	40%	0%
Final Exam	70%	50%	100%

The final mark will be the highest from **Option 1**, **Option 2** and **Option 3**.

Expectations:

(a) **Come to class.** Learning statistics is a cumulative experience; what we learn in one class builds upon what was covered in previous sessions. Missing class creates gaps in your knowledge base that will make hard to build on, and the degree of difficulty will increase rapidly.

(b) **Complete all assignments.** Much like the practice required in learning a foreign language, you will have to practice the techniques we learn in each class to get them to take hold and the homework is a good way to accomplish that. These exercises also will help you familiarize yourself with the software necessary to perform efficiently even simple analyses.

(c) **Ask questions.** Office hours provide a great opportunity to ask for clarifications of the material and questions about your own attempts to solve a problem. When you are stuck, try to explain someone else (classmate friend, tutor, instructor) what you are trying to do. This can often help you identify for yourself the problem you are having. Often the step we are missing is readily obvious to someone who is not trying to solve the problem.

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Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty please refer to the Academic Dishonesty Policy, located at `http://www.mcmaster.ca/academicintegrity`.

The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Abcenses: Please check

`http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work`

for the most up-to-date information about the **McMaster Student Absence Form** procedures (MSAF). Note that substantive changes were made this year to the procedures.

McMaster University

Department of Mathematics and Statistics

STATISTICS 4CI3/6CI3: Computational Methods for Inference

Winter 2016

Instructor	Angelo J. Canty
Office	Hamilton Hall 209
Phone	(905) 525-9140 Ext 27079
e-mail	cantya@mcmaster.ca
Course Web Site	http://www.math.mcmaster.ca/canty/teaching/stat4ci3
Time of Lectures	Monday 3:30 – 4:20
	Tuesday 4:30 – 5:20
	Thursday 3:30 – 4:20
Location	HH 217
Office Hours	Monday 2:30–3:30
	Thursday 2:30–3:30
	or at other times by prior appointment.
Text	There is no required text for this course but the material will be taken from the following texts
	Introducing Monte Carlo Methods with R
	C. Robert & G. Casella, Springer, 2010.
	Bootstrap Methods and Their Application
	A.C. Davison and D.V. Hinkley, Cambridge University Press, 1997.

Grading Scheme :

The final grade for undergraduate students (S4CI3) will be based on the following components

1. 40% for 4 assignments.
2. 30% for the midterm test.
3. 30% for the final exam.

The final grade for graduate students (S6CI3) will be based on the following components

1. 30% for 4 assignments.
2. 25% for the midterm test.
3. 25% for the final exam.
4. 20% for a course project.

Note that there will generally be an extra question on assignments for graduate students.

Notes :

- Students who miss work during the term for medical or other reasons should review and follow the Academic Regulation in the Undergraduate Calendar *"Requests for Relief for Missed Academic Term Work"*. Please note that these regulations have changed beginning Fall 2015. In particular, the online MSAF may not be used for your term test since it is worth more than 25% of your final grade so students will be required to provide the Associate Dean with appropriate documentation in the event of a missed test and also to see the instructor to discuss any possible accommodation. For missed assignments the MSAF system may be used once and the total assignment grade will be calculated based on the other 3 assignments.

- Tentative due dates for the assignments will be at **the start of class** on Feb. 4, Feb. 25, Mar. 17, and Apr. 7. Assignments will be posted on the web site at least 1 week before the due date.

- The midterm test will be held during class time. Tentative date for the test is March 3. The test will take place in the usual lecture room during regular class time. The exact coverage of the test will be announced in class.

- The final exam will take place during the April examination period and will cover the entire course. It will be a 2.5 hour exam common to both graduate and undergraduate students.

- The project for Stat 6CI3 will require students to conduct a simulation study and to write a report describing the study and the results. Students are required to submit a short (1 page) proposal prior to starting their project. The final date for submission of the proposal is March 24, 2016. The final report should be about 10 pages and be formatted using LaTeX. It should be submitted electronically accompanied by a file of commented R code used by April 15, 2016.

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Academic Dishonesty

Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3, located at http://www.mcmaster.ca/senate/academic/ac_integrity.htm

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- **Improper collaboration in group work.**
- **Copying or using unauthorized aids in tests and examinations**

Brief Outline of the Course

This course is designed to introduce students to important computational techniques used in modern statistical inference. Students will also be introduced to programming in R and all computation in this course is expected to be done in this language. R may be downloaded free of charge from https://cran.r-project.org for Windows, Macintosh or Linux operating systems. Students are not required to know R but may find it helpful to consult some of the available online documentation such as

- *An Introduction to R* by W.N. Venables, D.M. Smith and the R Core Team.
- *R for Beginners* by Emmanuel Paradis.
- There is also much more useful documentation available at `http://cran.r-project.org/other-docs.html`
- There are also books on R (or S on which R is based) programming available in the Thode library.

The topics to be covered in the course will be

1. Fundamentals of programming in R;
2. Computer generation of random variates;
3. The Monte Carlo method and simulation;
4. Introduction to simulation studies;
5. Markov chain Monte Carlo methods;
6. Jackknife and bootstrap estimation of bias and variance;
7. Monte Carlo hypothesis testing;
8. Bootstrap confidence intervals.

Department of Mathematics & Statistics,McMaster University

Statistics 4D03/6D03: Intermediate Probability Theory

Course Outline Winter: 2015/2016

Instructor: Dr. S. Feng, HH/211, Extension 24723

E-mail: shuifeng@mcmaster.ca

Lectures: Tu, We, and Fr 12:30 – 13:20 BSB/104

Office Hours: Fridays 1:30pm-3:30pm

Main Reference: Probability: Theory and Examples (all editions) by Richard Durrett.

Course Web: http://ms.mcmaster.ca/~fengs/S4D03-15.html

Course Contents: probability space, random variables, types of convergence, weak and strong law of large numbers, expectation and conditional expectation, characteristic function, continuity theorem, central limit theorem, infinitely divisible laws, martingales.

Tests, Assignments and Exam: There are five assignments, two tests and a final exam.

Marking Scheme:	Assignments	20% (4% each)
	Tests	30% (15% each)
	Exam	50%

Test dates (in class):	Test #1:	Friday, Feb. 5, 2016 (tentative)
	Test #2:	Friday, Mar. 18, 2016 (tentative)

Calculators: The only calculator allowed for the tests and final exam will be a Casio fx-991.

The instructor and university reserves the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Policy regarding missed work:If you have missed work, it is your responsibility to take action.

Details concerning the missed term work and up-to-date MSAF information, please refer to the following urls:
https://www.mcmaster.ca/msaf/index.html

http://academiccalendars.romcmaster.ca/content.php?catoid=13&navoid=2208#Requests_for_Relief_for_Missed_Academic_Term_Work

Please note that the MSAF may not be used for term work worth 25% or more,nor can it be used for the final examination.

ACADEMIC DISHONESTY

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It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of a academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3, located at http://www.mcmaster.ca/senate/academic/ac_integrity.htm

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2. Improper collaboration in group work.

3. Copy or using unauthorized aids in tests and examinations.

Course Outline · STATS 4M03/6M03 · Fall 2015

Course Material
Course material can be accessed via the appropriate link at the teaching section of my website `http://ms.mcmaster.ca/~paul`. You will need a username and password to access the material; this has already been sent by email and should not be circulated beyond students in this class.

Instructor: Dr. Paul McNicholas, HH210, `mcnicholas@math.mcmaster.ca`

Office hours: To be announced.

Course meeting times: Tuesday, Thursday, and Friday at 11:30AM in HH 217.

Textbook: There is no textbook for this course.

Course structure: Three lectures per week.

Course objective
To learn the fundamentals of multivariate analysis. To understand how to use and interpret results from some of the techniques commonly used for multivariate data.

Assessment

- Assignments: 30%. There will be two or three assignments. Each assignment will be available at least one week before it is due.
- Midterm exam: 35%. There will be one midterm exam. The tentative date for the midterm exam is October 8.
- Project: 35%. There will be one project, which may include an oral presentation thereof. The tentative due date for the project (write-up and code) is November 27. Oral presentations, if used, will take place thereafter.

MSAF (only applies to undergraduate students, i.e., students in STATS 4M03)
In the event of an absence for medical or other reasons, students in STATS 4M03 should review and follow the regulations in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

Academic Accommodation of Students with Disabilities
Students who require academic accommodation must contact Student Accessibility Services (SAS) to make arrangements with a Program Coordinator. Academic accommodations must be arranged for each term of study. Student Accessibility Services can be contacted by phone 905-525-9140 ext. 28652 or email `sas@mcmaster.ca`. For further information, consult McMaster University's Policy for Academic Accommodation of Students with Disabilities.

McMaster Policy on Academic Integrity

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Please Note

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of students to check **their McMaster email** weekly during the term and to note any changes. Announcements will be made in class and via email.

Topics Covered in STATS 4M03

Multivariate random variables
Expectation, variance, covariance
Principal components
Multivariate normal distribution
Factor analysis
EM algorithm
Classification and discriminant analysis
Cluster analysis
Binary logistic regression
Multinomial logistic regression
Hotelling's T2
Hypothesis test and confidence region for multivariate mean
MANOVA
Canonical correlation
Copulas
Classification trees
ROC curves
Regression trees

STAT 4P03/6P03: Advanced Applied Statistics.
Winter 2015 (Term 2)

Instructor:
Brian Franczak, PhD.
Hamilton Hall Room 208,
Department of Mathematics and Statistics,
McMaster University.
Email: bfrancza@math.mcmaster.ca

Lecture Times and Location:
Lecture: Tuesday, Thursday, and Friday from 11:30am to 12:20pm
Location: John Hodgins Engineering (JHE) Building 329

Teaching Assistant (TA):
Yang Tang

Office Hours:
Monday and Thursday 1:30 - 2:30pm

Course Materials:
Textbooks:

- Chatfield, C. (1988). *Problem Solving: A Statisticians Guide*. Chapman and Hall.
- Hastie, T., R. Tibshirani, and J. Friedman (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.
- Everitt, B.S., S. Landau, M. Leese, and D. Stal (2011). *Cluster Analysis* (2nd ed.). John Wiley & Sons, Ltd.
- Johnson, R.A. (1992). *Applied multivariate statistical analysis*. NJ: Prentice hall*.
- Little, R.J. and D. B. Rubin (2002). *Statistical Analysis with Missing Data* (2nd ed.). Wiley.
- van Buuren, S. (2012). *Flexible Imputation of Missing Data*. CRC Press.

Note: * indicates that this book is not available online

Journal Articles:

- Will be provided.

Software:

- R: The R Project for Statistical Computing (Freeware available online at `www.r-project.org`)
- LaTeX: A document preparation system (Freeware available online at `www.latex-project.org`)

Course Website:
Avenue to Learn

Evaluation:
Four assignments worth 50% (12.5% each)
One final project worth 45%
Attendance and Participation 5%

Assignments:
The following is a tentative schedule of dates. You will have two weeks to complete each assignment. Submit them directly to me at the beginning of lecture on the dates listed below.

Assignment	Due Date
1	Friday, January 30, 2015
2	Friday, February 13, 2015
3	Tuesday, March 10, 2015
4	Tuesday, March 31, 2015

Final Project:
You will have one month to complete the final project. Submit it directly to me in Hamilton Hall Room 208 between 8:00am and 3:30pm on the tentative date listed below.

Final Project	Due Tuesday, April 28, 2015

Course Objectives:
1. Develop your research and communication skills (both oral and written).
2. Make you familiar with (and capable of using) R, SAS, and LaTeX.
3. Make you aware of your post-undergraduate opportunities.

Syllabus:
1. Approaches to Data Analysis.
   a. A general strategy.
2. Generalized Linear Models.
   a. Model fitting and interpretation.
3. Multivariate and "Big Data."
   a. Definitions and nomenclature.
4. Dimension Reduction Techniques
   a. Principal Component Analysis
   b. Factor Analysis
5. Cluster Analysis
   a. Proximity clustering
   b. Hierarchical clustering
   c. Model-based approaches
6. Imputation
   a. Single vs. Multiple Imputation
7. Other Topics
   a. To be determined

Email Ediquette:
I may not respond to emails asking for "trivial" information that has been provided in class or is easy to find on the course website. I am going to focus on addressing concerns with the course material...

Academic Integrity:
From McMaster University's Policies, Procedures and Guidelines Academic Integrity Policy:

> "Students are responsible for being aware of and demonstrating behaviour that is honest and ethical in their academic work."
>
> "Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage"

As your instructor I am responsible for submitting a statement of any offences you commit to the appropriate governing body.

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MSAF:
MSAF information link:
`http://academiccalendars.romcmaster.ca/content.php?catoid=7&navoid=559#Requests_for_Relief_for_Missed_Academic_Term_Work`

Regarding missed term work:
`https://www.mcmaster.ca/msaf/index.html`